



Technical Training

Air Conditioning System Diagnostics

TRAINERS: Wilson Almeida



ACADÉMIE
Vast auto

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Training objective

What this training covers

By the end of the day, you will be able to:

- Explain the fundamental principles of air conditioning
- Explain the components of an air conditioning system
- Explain the work methodology for an A/C system
- Explain the diagnostic approach for an A/C system

Halocarbons

Item	Canada – Federal	Québec – Provincial
Regulation	Federal Halocarbon Regulations (2022)	Halocarbons Regulation (Q-2, r.29)
Halocarbon release	✗ Prohibited	✗ Prohibited
Recovery mandatory	✓ Yes (servicing and decommissioning)	✓ Yes (servicing and decommissioning)
Leak testing	✓ Required	✓ Required
Qualified personnel	✓ Mandatory (certification)	✓ Mandatory (certification)
Recordkeeping	✓ Mandatory	✓ Mandatory
Main objective	Reducing emissions and protecting the climate	Protecting the environment and the ozone layer

Requirement	Example of offence	Type	Technician	Company	Sections
 Certification	Not carrying or showing the certification	AMP	250 \$	1 000 \$	61.1 (2°)
 Certification	Working without qualification	AMP	500 \$	2 500 \$	61.3
 Certification	Working without certification	PENAL	1 000 \$ to 100 000 \$	3 000 \$ to 600 000 \$	46 → 62
 Release into the air	Releasing halocarbon into the atmosphere	AMP	2 000 \$	10 000 \$	61.7
 Release into the air	Uncontrolled release or leak	PENAL	12 500 \$ to 1 000 000 \$	37 500 \$ to 6 000 000 \$	5 → 67.2
 Leak test	Test not performed	AMP	500 \$	2 500 \$	61.3 (1°)
 Leak test	Repeated or serious failure	PENAL	2 500 \$ to 250 000 \$	7 500 \$ to 1 500 000 \$	9 → 64
 Logbook	Logbook missing or not kept	AMP	250 \$	1 000 \$	61.1 (4°)
 Logbook	Logbook not maintained or not submitted	AMP	350 \$	1 500 \$	61.2
 Logbook	Serious or repeated breach	PENAL	2 000 \$ to 100 000 \$	6 000 \$ to 600 000 \$	59 → 63

⌚ **64.** Every person who

- (1) contravenes section 7, the first or second paragraph of section 9, section 22, 43, 50 or 51,
- (2) fails to have an assessment made of the quantity of halocarbon released during a leak, in accordance with the second paragraph of section 11,

commits an offence and is liable, in the case of a natural person, to a fine of \$2,500 to \$250,000 or, in other cases, to a fine of \$7,500 to \$1,500,000.

O.C. 1091-2004, s. 64; O.C. 676-2013, s. 8; O.C. 201-2020, s. 67.

⌚ **9.** A person who fills or refills a container or a refrigeration or air conditioning unit with a halocarbon or charges or recharges a fire extinguisher is required, subject to section 12, to first leak test the equipment.

No person may use sulfur hexafluoride (SF₆) to conduct the leak test referred to in the first paragraph.

O.C. 1091-2004, s. 9; O.C. 201-2020, s. 7.

<https://www.legisquebec.gouv.qc.ca/en/document/cr/q-2,%20r.%2029>



**Use only electronic leak detectors that meet SAE J2913
for use with R-1234yf**

Vehicle air conditioning unit or transport refrigeration unit

1. Identification			
NAME OF THE WORKER:			
ENVIRONMENTAL QUALIFICATION CERTIFICATE NUMBER:			
NAME OF THE EMPLOYER:			
ADDRESS:			
CITY:		POSTAL CODE:	

2. Type of unit	
TYPE OF VEHICLE:	
VEHICLE DESCRIPTION (make, model and year):	
SERIAL NUMBER:	
LICENCE PLATE NUMBER:	

3. Halocarbon recovery (if applicable)			
A	LEAK TEST RESULT OF THE RECOVERY CONTAINER:	☐ PASSED	☐ FAILED
	HALOCARBON RECOVERED:	TYPE:	QUANTITY (kg)
B	IS THE RECOVERED HALOCARBON KEPT BY THE OWNER? If yes, complete section C	☐ YES	☐ NO
	NAME OF THE OWNER:		
C	ADDRESS:		
	CITY:	POSTAL CODE:	

4. Nature of the work			
A	SUMMARY DESCRIPTION OF THE WORK:		
	Addition of a halocarbon (if applicable)		
B	LEAK TEST RESULT:	☐ PASSED	☐ FAILED
	HALOCARBON ADDED:	TYPE:	QUANTITY (kg)

I certify that the information provided in this report is accurate:

Name of the authorized person (block letters):	
Signature of the authorized person:	
Date:	

5 years / 2 copies

System identification

What each standard covers

The three SAE standards to know

- J639: safety procedures for pressurized air conditioning systems
- J2842: evaporator design
- J2845: repair standards and equipment to be used



Let us compare the refrigerants

	R134a		R1234yf	
Chemical name	tetrafluoroethane		tetrafluoropropene	
Chemical formula	CH_2FCF_3		$\text{CF}_3\text{CF}=\text{CH}_2$	
Recommended oil	Polyalkylene Glycol (PAG) (polyol ester – POE)		Polyalkylene Glycol (PAG) designed for R1234yf (polyol ester – POE)	
ASHRAE classification	A1 – non-toxic, non-flammable		A2L – non-toxic, mildly flammable	
Boiling point at 1 atm	-26 °C		-29 °C	
ODP	0		0	
GWP	1430		4	
GWP as a share in % of R134a	100 %		0.3 %	

Characteristic	R-134a	R-1234yf	R-744
Chemical type	HFC	HFO	CO ₂ (carbon dioxide)
Use	Vehicles ~1994–2024	Transition starting in 2011 Mandatory as of 2024	New high-pressure systems
GWP	≈ 1430	≈ 4	1
Flammability	Non-flammable	Mildly flammable	Non-flammable
Operating pressure	Medium	Medium	Very high
Typical high-side pressure	150–250 psi	140–240 psi	1000–2000 psi
Energy efficiency	Good	Very similar to R-134a	Very good under certain conditions
Oil used	PAG, POE	PAG, POE, specific PVE	POE oil / CO ₂ -specific
Recharging machine	Standard R-134a	Dedicated R-1234yf machine	High-pressure CO ₂ machine
Service fittings	Standard R-134a	Different, for safety	Different, high pressure
Environmental impact	High	Very low	Very low
Regulatory status	Being phased out	Current standard	Emerging technology

New fittings

Why the fittings change

- New fittings to prevent interchangeability
- Different dimensions, longer and smaller



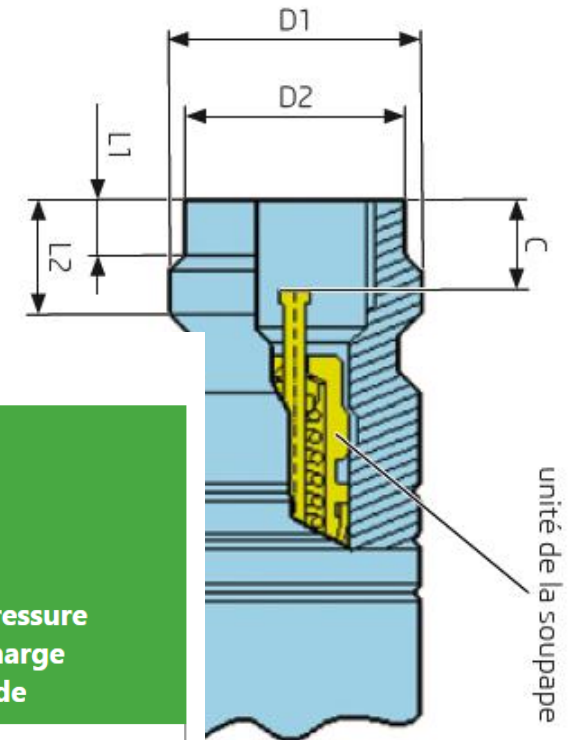
R1234yf



R134a



Fitting dimensions

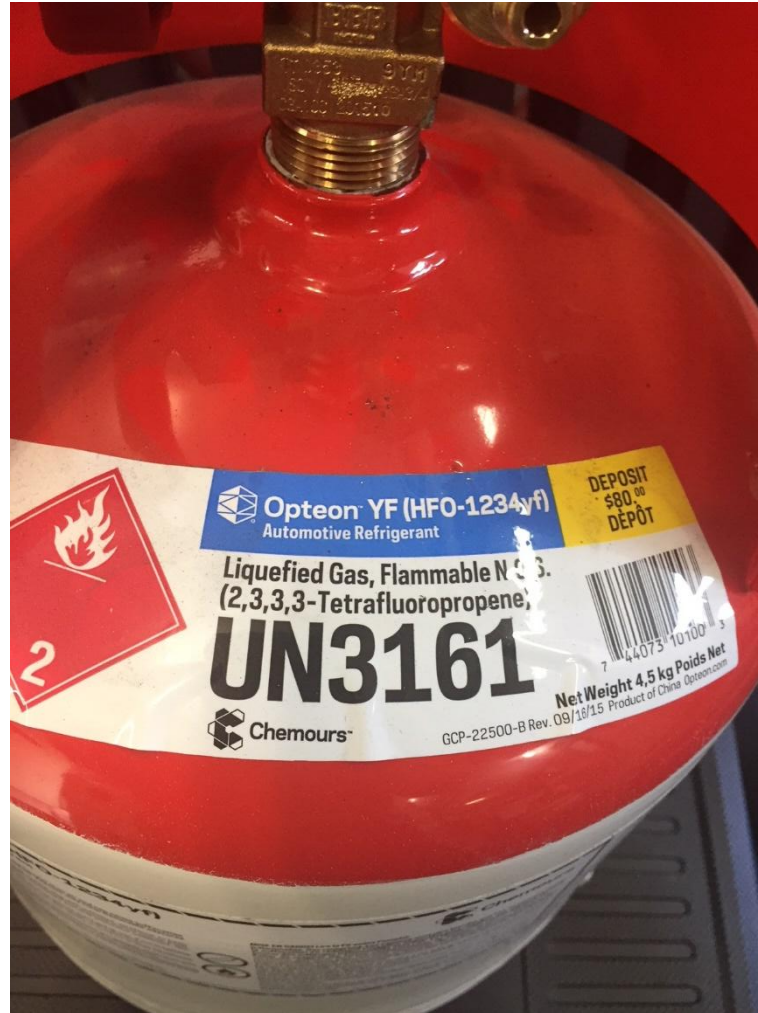


The table describes the parameters of the four non-interchangeable service fittings:

Service fitting dimensions	Service fitting, refrigerant R134a		Service fitting, refrigerant R1234yf	
	high-pressure discharge side	low-pressure discharge side	high-pressure discharge side	low-pressure discharge side
outside diameter (D1)	16.0 mm	13.0 mm	17.0 mm	14.0 mm
outside diameter (D2)	14.0 mm	11.0 mm	13.0 mm	12.0 mm
fitting length (L1)	4.6 mm	6.15 mm	9.0 mm	4.75 mm
fitting length (L2)	8.16 mm	9.16 mm	12.5 mm	7.2 mm
valve mounting position, not connected (C)	6.1 – 7.1 mm	6.1 – 7.1 mm	8.3 – 9.3 mm	8.3 – 9.3 mm

The R1234yf refrigerant container

- Standardized white and red container
- Fitted with a one-way valve



No retrofitting, please!

The rule

- Retrofitting a refrigerant is prohibited, in either direction
- R134a refrigerant remains available for the vehicles that use it

⚠ WARNING Even if you know that some people still do it: retrofitting remains prohibited, in both directions.

The fines set out in the regulation

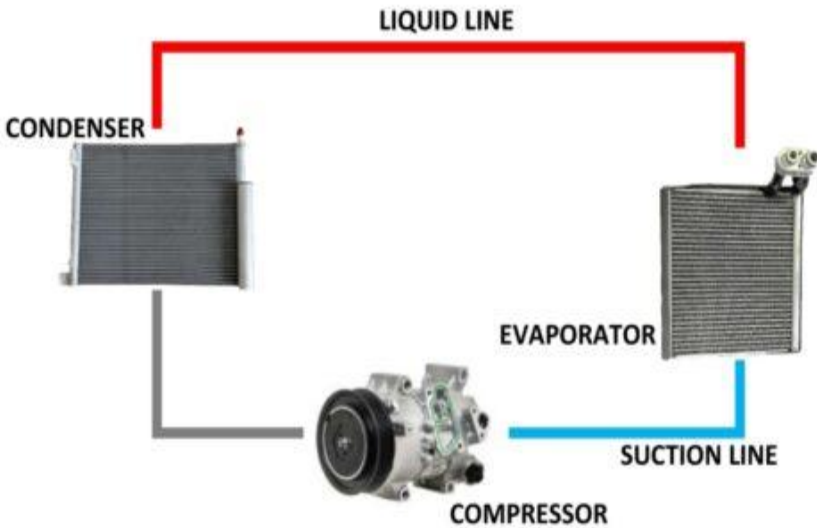
Type of offence	Section of the regulation	Technician (individual)	Company
Modifying or converting a unit in breach of the regulation	S. 64	2 500 \$ to 250 000 \$	7 500 \$ to 1 500 000 \$
Using a non-compliant or unauthorized refrigerant in a unit	S. 67.3	1 000 \$ to 100 000 \$	3 000 \$ to 600 000 \$

The changes

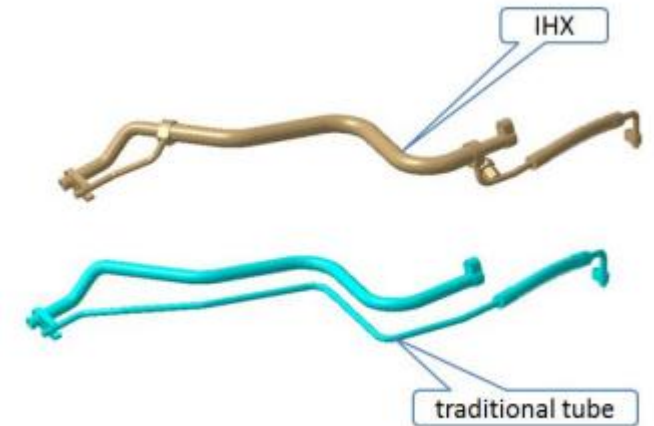
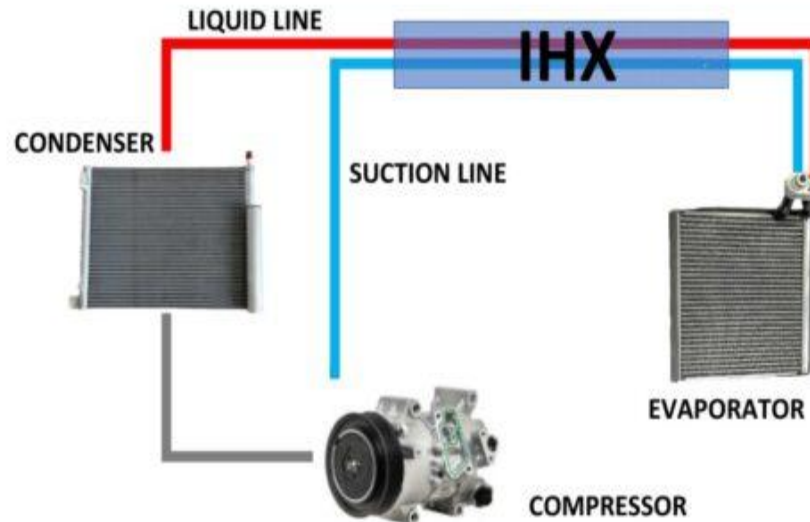
What changes with R1234yf

- Similar to the changes in the R12 to R134a transition
- Differences in the service fittings
- Slight differences in the expansion valve, the compressor and the condenser
- Modified evaporator, which must meet SAE J2842
- Addition of an internal heat exchanger, but NOT ON ALL VEHICLES

CONVENTIONAL AC SYSTEM



AC SYSTEM WITH IHX

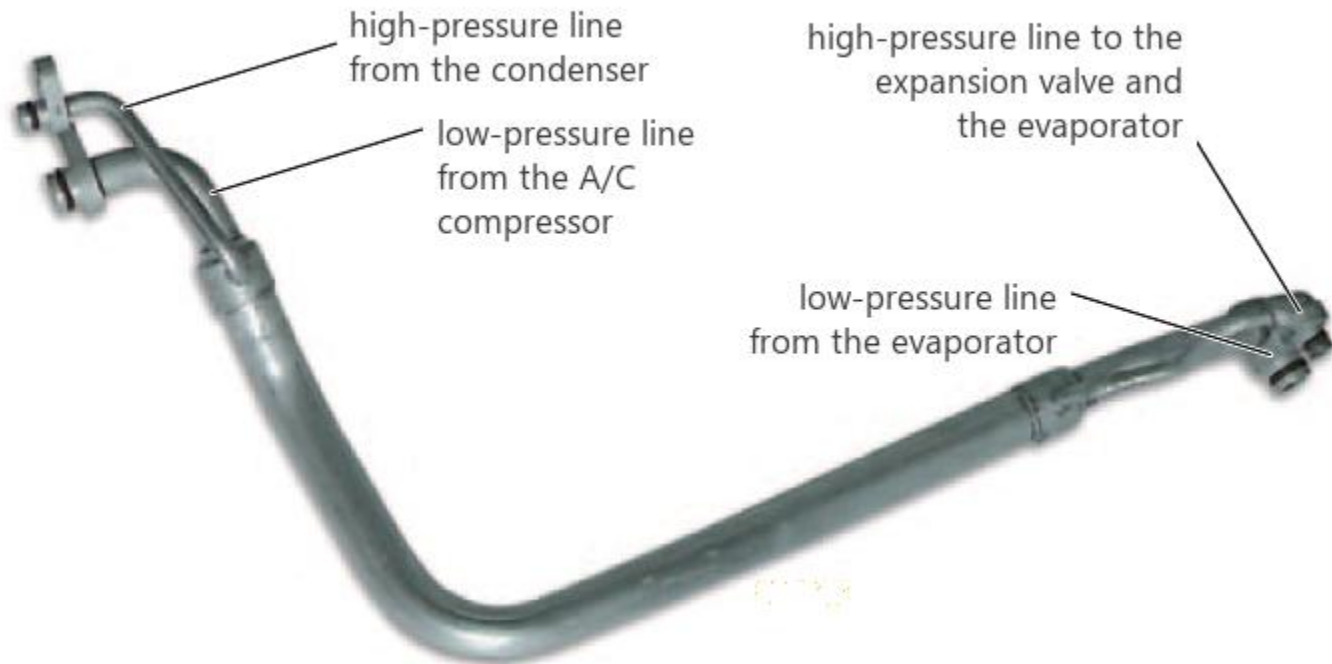
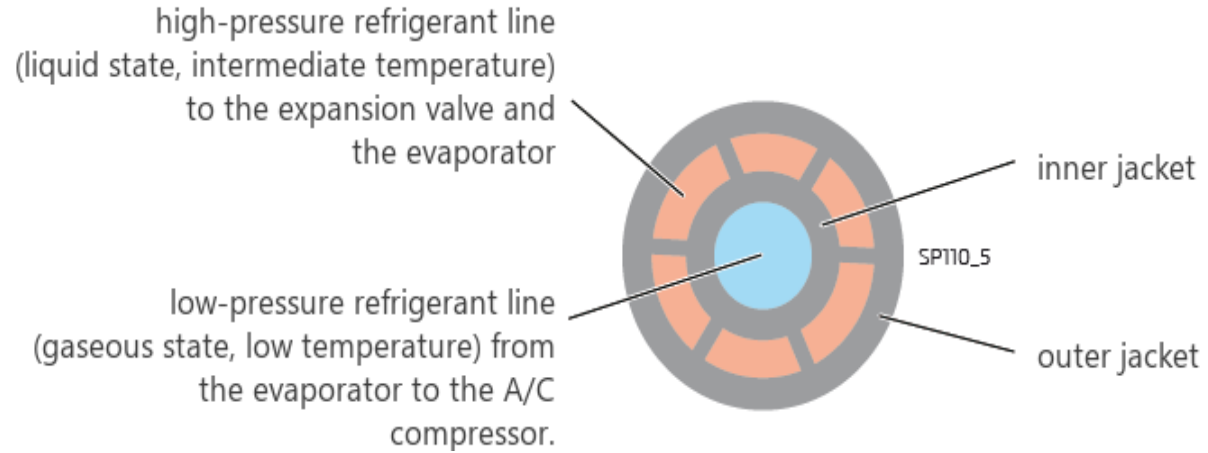


Role of the internal heat exchanger

- Provides greater efficiency in condenser subcooling
- Ensures the refrigerant converts from gas to liquid
- Used on some vehicles that still use R134a
- Not used on all vehicles that use R1234yf

How does it work?

Cross-section of the internal heat exchanger



How does it work?



How the internal heat exchanger is built

- A treated aluminum pipe with two jackets
- Outer cavity: the high-pressure liquid refrigerant entering the evaporator
- Inner cavity: the low-pressure gaseous refrigerant leaving the evaporator
- The high-pressure liquid is cooled by the low-pressure gas that surrounds it

The regulations

What the law requires, R1234yf included

- R1234yf refrigerant contains halocarbons
- The regulations remain unchanged in Canada and in Québec
- The technician must hold a certification to handle halocarbons
- Diagnostic, repair and safety procedures must comply with SAE J639 and J2845
- The equipment used must meet the applicable SAE standards

► **What standard J2842 sets out**

- Product and process design requirements
- Validation tests and acceptance criteria for R-1234yf and R-744 evaporators

► **Design service life adopted**

- 10 years and 6 000 hours of operation
- Many manufacturers interpret this term differently depending on local climate conditions



A quick review of definitions

Heat and temperature

Air conditioning does not create cold

- It moves heat from the cabin air to the outside
- When heat decreases in the cabin, the air temperature drops as well
- Cooling the cabin means transferring heat from the cabin to the outside

Heat and temperature are not the same thing

- Heat is thermal energy, it is measured in BTU
- Temperature is a thermal level, it is measured in °C
- The two terms are not interchangeable



BTU: 40 000
Max temperature: 600°C



BTU: 100 000
Max temperature: 600°C



Temperature: 1200°C
Several days
1 BTU



Temperature: 3°C
A few hours
250 BTU



Temperature: 20°C
30 minutes
400 BTU

A quick review of definitions

The two forms of heat

Sensible heat

- Heat exchanged with no change of state
- It changes the temperature of the body, so it can be felt and measured
- That is why it is called « sensible »

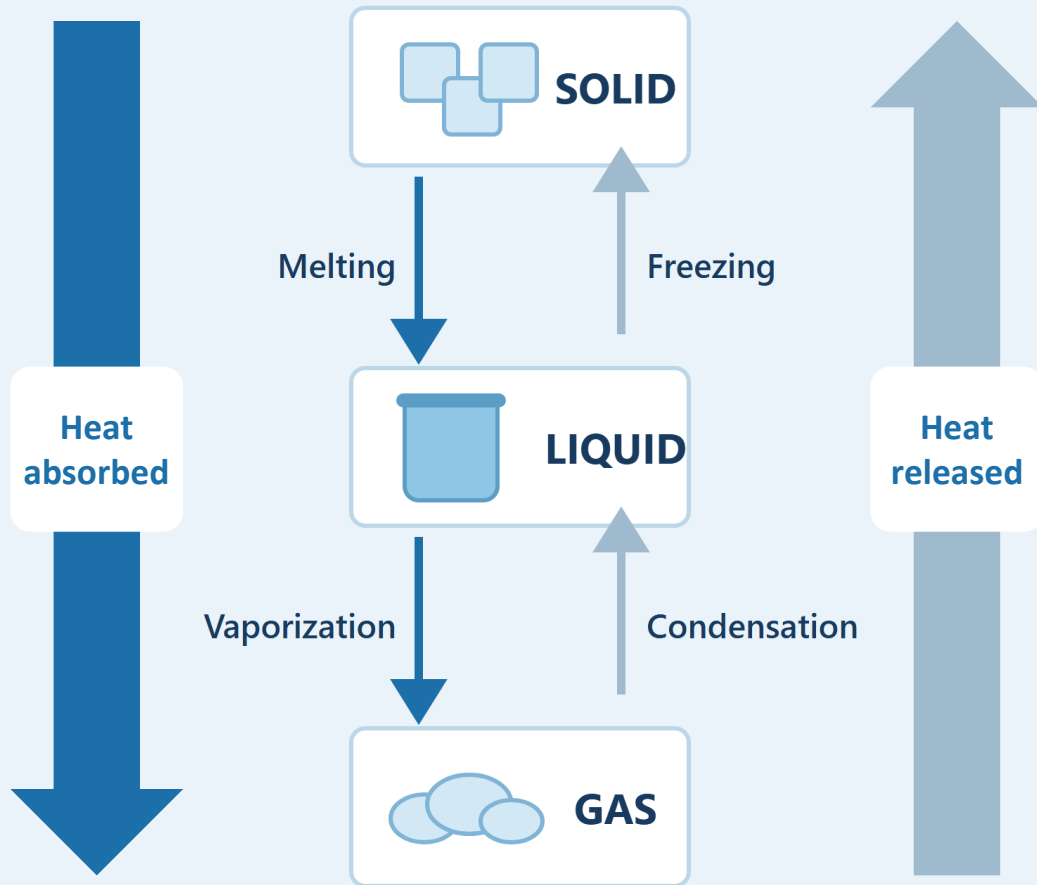
Latent heat

- Heat absorbed or released during a change of state
- It does not change the temperature, only the state of the body
- Example: going from the liquid state to the vapour state, vaporization

 **KEY POINT** Latent heat does all the work in an A/C system: the refrigerant absorbs a huge amount of heat as it goes from liquid to vapour, with no change in its temperature.

Latent heat

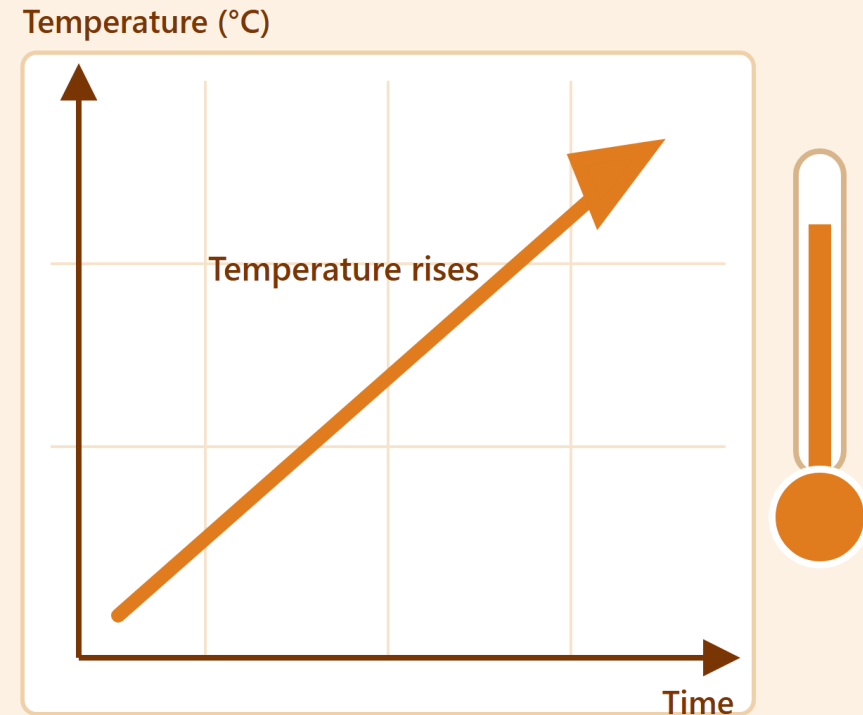
the state changes, the temperature does not



Heat is absorbed or released
with NO temperature change

Sensible heat

the temperature changes, you can measure it



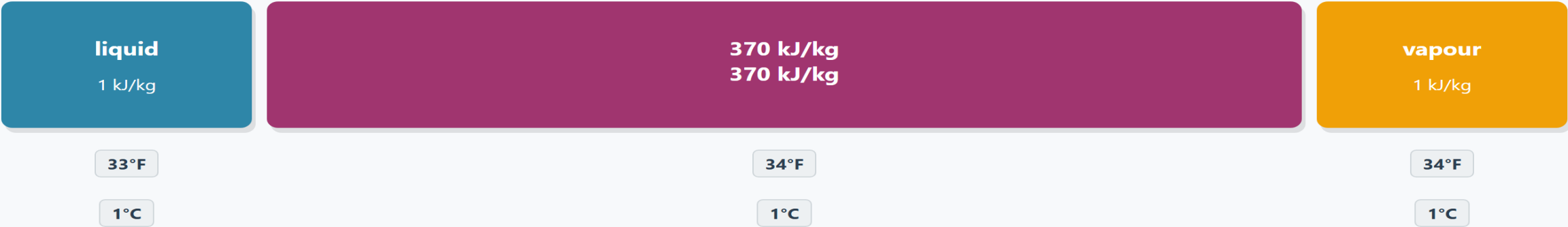
Heat is exchanged
and the TEMPERATURE CHANGES

P-T diagrams: change of state

WATER at 0 PSI



R134a at 30 PSI





Friction = sensible heat

- Rubbing warms the skin slightly
- That brings energy to the alcohol
- Up to this point the temperature rises a little: this is sensible heat

Evaporation = latent heat, the key point

- Alcohol has a low boiling point
- It goes from liquid to vapour quickly
- To change state it needs energy, and that energy is taken from your skin

 **KEY POINT** Alcohol cools the skin because it absorbs the latent heat needed for it to evaporate, not because it is cold to begin with.

A quick review of definitions

The two basic measurements in diagnosis

Evaporator superheat

– INCREASE in the refrigerant temperature

Condenser subcooling

– DECREASE in the refrigerant temperature

PSIG	°F
16	16
18	19
20	22
22	25
24	27
26	30
28	32
30	34
32	36
34	39
36	41
38	44
40	48
42	47
44	49
46	50
48	52
50	54
55	58
60	62
65	66
70	69
75	73
80	76
85	79
90	82
95	85
100	88
105	91
110	93
115	96
120	98

EVAPORATOR

PSIG	°F
125	101
130	103
135	105
140	107
145	110
150	112
160	116
170	120
180	124
190	127
200	131
210	134
220	137
230	141
240	144
250	147
260	150
270	152
280	155
290	158
300	160
310	163
320	165
330	168
340	170
350	172
360	175
370	177
380	179
390	181
400	183
410	185

CONDENSER

R-134a

Fahrenheit

Pressure/ Temperature Chart

Torque Settings

#6 O-Ring Fitting: **11-13 ft-lb**

#8 O-Ring Fitting: **15-20 ft-lb**

#10 O-Ring Fitting: **21-27 ft-lb**

#12 O-Ring Fitting: **28-30 ft-lb**

#14 O-Ring Fitting: **30-35 ft-lb**

Pressure switch on Schrader
port (7/16-20): **85-90 in-lb**

Pressure switch
on M10 port: **60-65 in-lb**

Schrader port
valve insert: **3-5 in-lb**

R-1234yf

Fahrenheit

Pressure/ Temperature Chart

PSIG	°F
9	0
11	4
14	8
16	12
19	16
19	17
20	18
23	22
24	23
25	24
28	28
31	32
33	34
35	36
37	38
38	40
40	42
42	44
44	46
47	48
49	50
51	52
53	54
56	56
58	58
61	60
63	62
66	64
68	66
71	68
74	70

EVAPORATOR

PSIG	°F
77	72
80	74
83	76
86	78
89	80
92	82
96	84
99	86
102	88
106	90
110	92
113	94
117	96
121	98
125	100
129	102
133	104
137	106
142	108
146	110
148	111
150	112
155	114
160	116
164	118
169	120
174	122
179	124
184	126
190	128
195	130

PSIG	°F
200	132
206	134
212	136
218	138
223	140
229	142
236	144
242	146
248	148
255	150
261	152
268	154
275	156
282	158
289	160
296	162
304	164
311	166
319	168
326	170
334	172
342	174
351	176
359	178
368	180
376	182
385	184
394	186
403	188
413	190

CONDENSER

Important points before starting

The inspection sequence before hooking up any gauges

Inspecting and understanding the system

- Basic inspection
- Understand how the system works (theory and operation)
- Check the condenser
- Check the compressor

Electronic and documentary checks

- Do we have any DTCs? Auto-scan
- TSBs and reflashes?

Physical checks

- Are any components missing?
- Evaporator drain: what colour is it?
- Do we have proper static pressures (pressure = temperature)

 **TIP** Static pressure is read after 10 minutes at rest: low and high sides must be equal and must match the ambient temperature. Careful, a correct static pressure does not prove the system is full, only that it holds 30 % or more of its charge.

Checking the air blend door

A leaking door throws off every diagnosis that follows

Test conditions

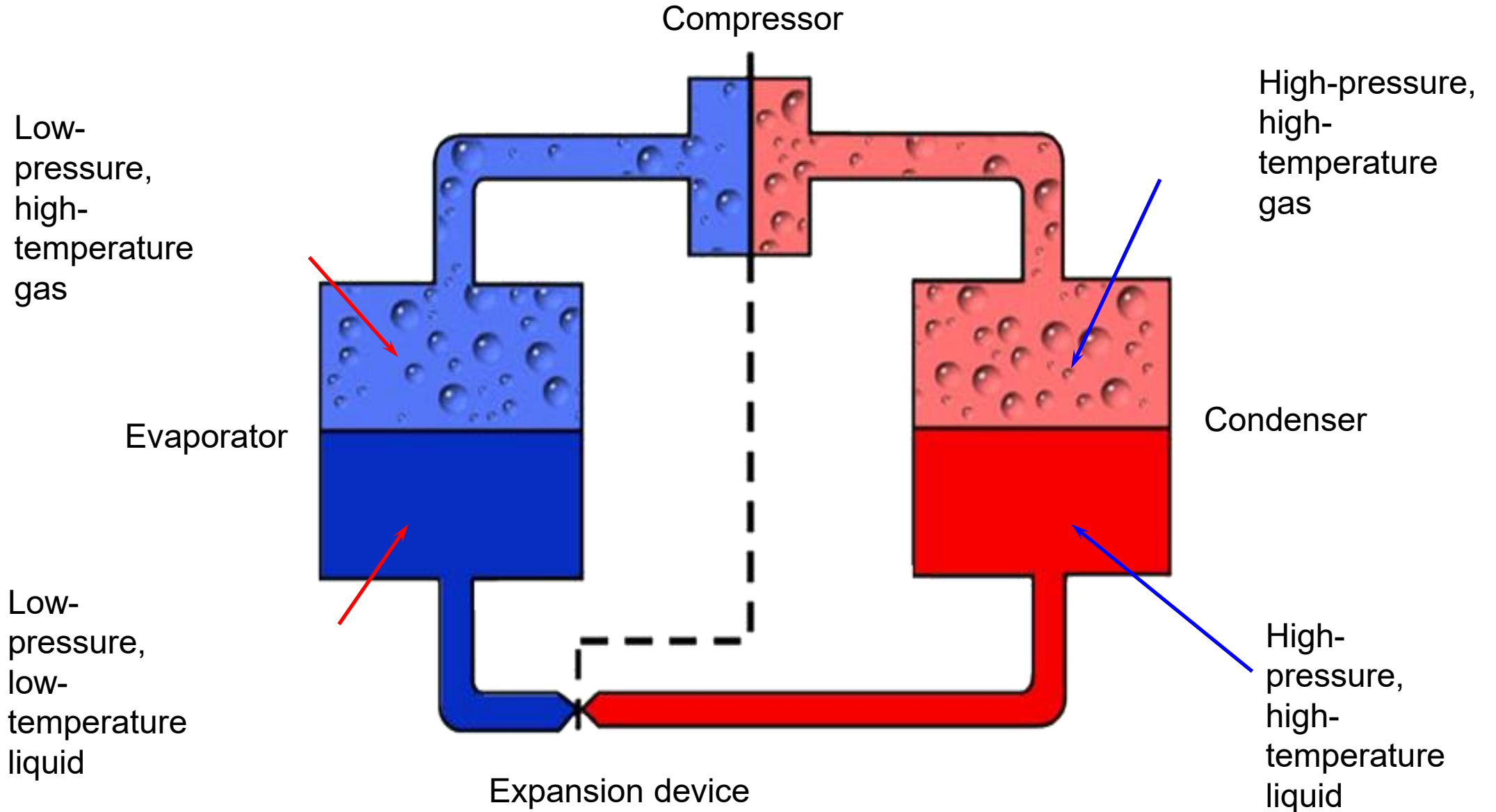
- A/C off
- Engine at operating temperature
- Hood closed, doors and windows open
- Control set to maximum cold
- Test each zone independently

Reading and specification

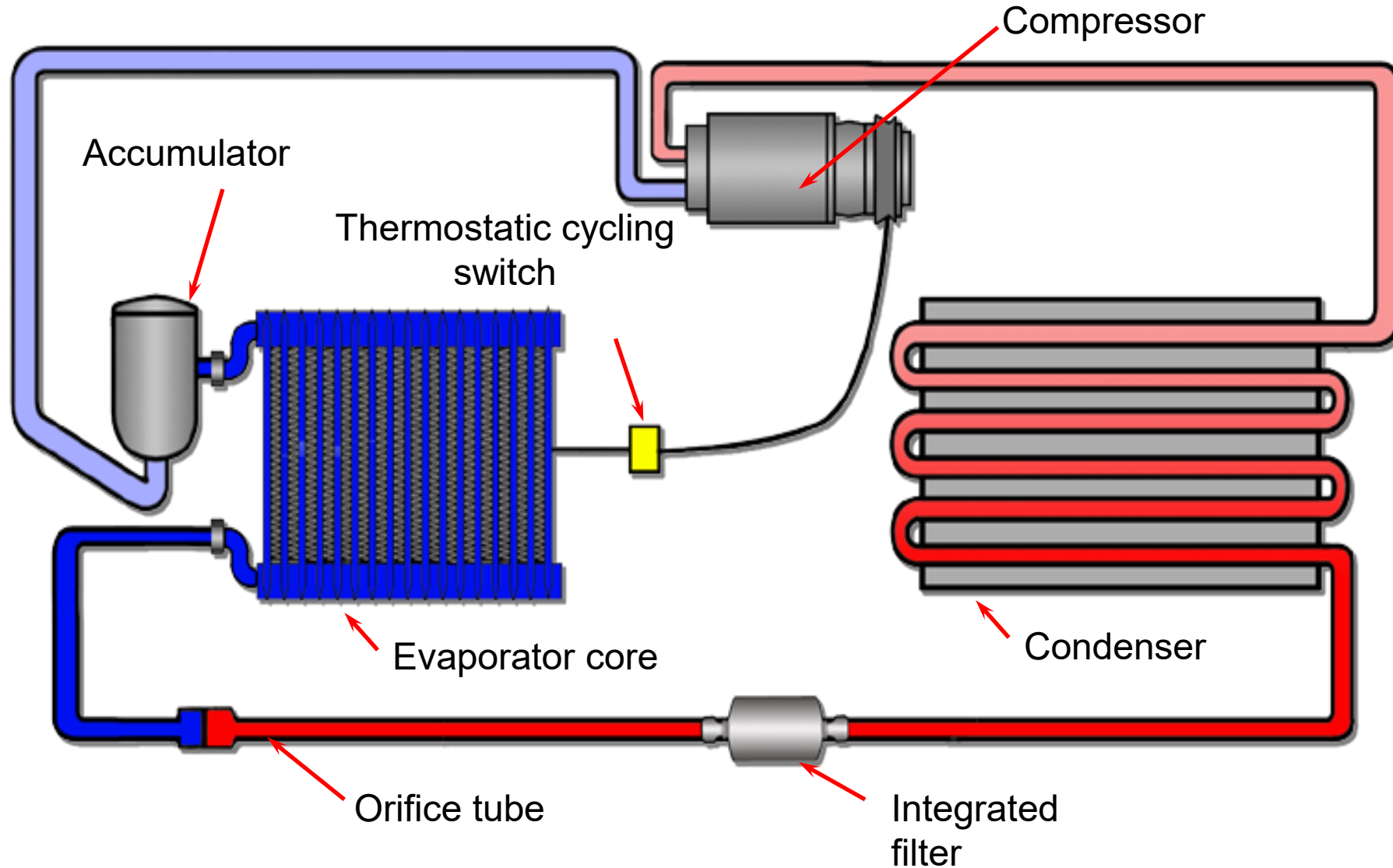
- Compare with the ambient temperature
- Spec: 5 °F difference between the outside temperature and the centre vent

⚠ WARNING If the difference is more than 5 °F, there is a door problem. It must be repaired BEFORE going any further with the diagnosis of the refrigerant circuit.

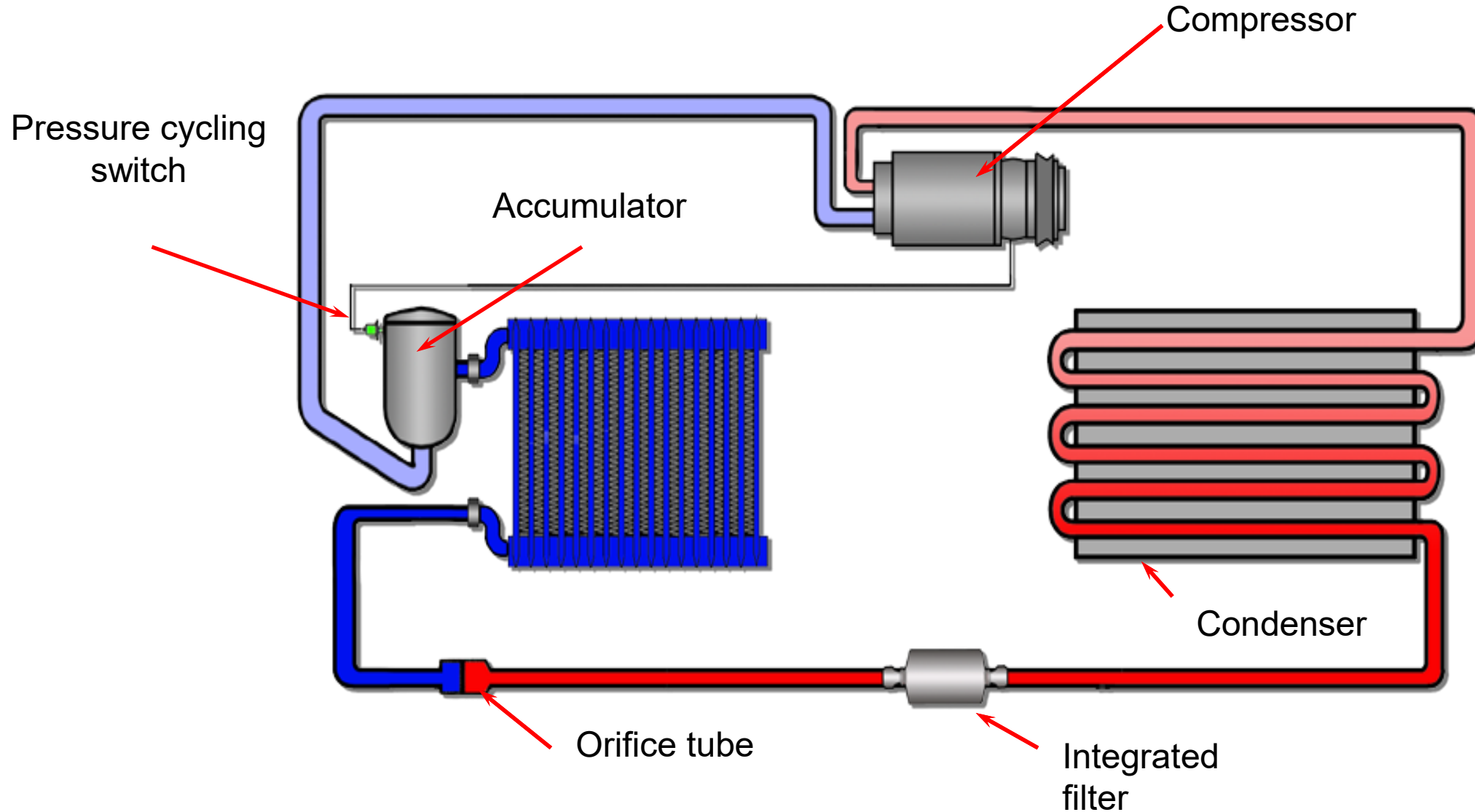
Applying the principles of physics



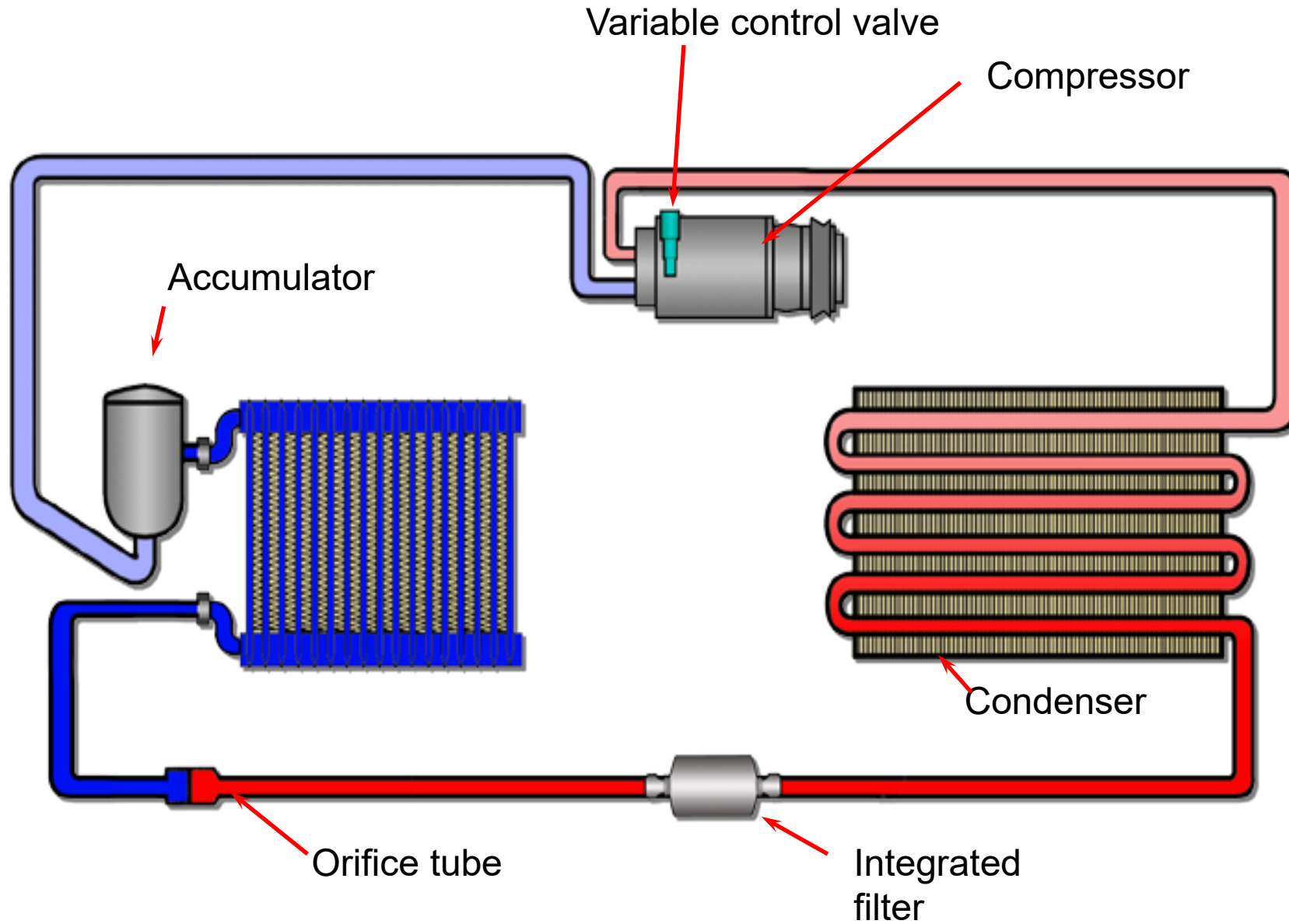
Thermostatic cycling system



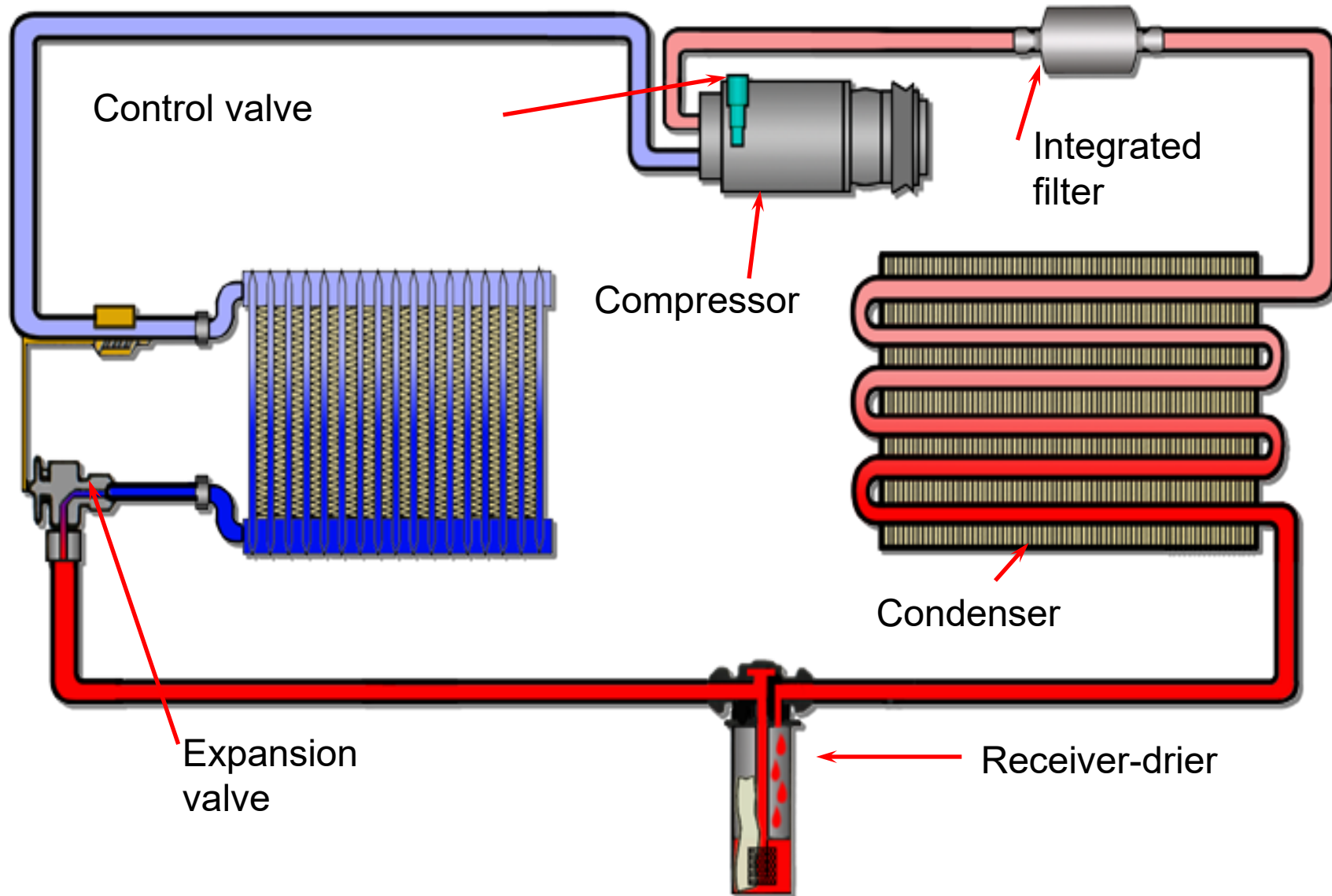
Pressure cycling system



VDOT system



VDTXV system



Very important when it comes to compressors

The oil quantity is a specification, not an estimate

Before reinstalling a compressor

- Add the correct amount of oil by draining and refilling it to specification
- Check the specification in Mitchell or at www.4s.com

Compressor failure

The most common causes

- Lack of lubrication
- Low refrigerant level
- Internal restrictions
- Contamination

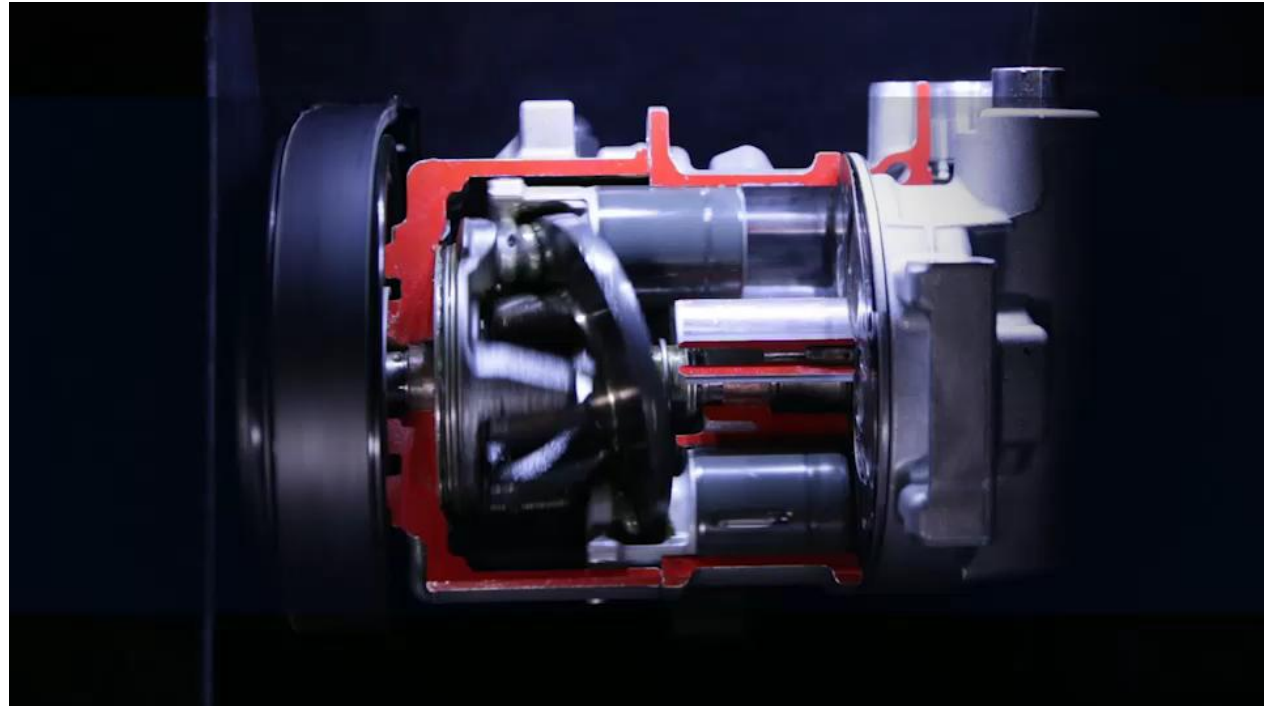


⚠ WARNING After a compressor failure, every inch of refrigerant passage should be new or cleaned. Otherwise the debris goes right back into the new system.

Variable compressor

Symptoms when the control valve sticks

- Pressure stays at the minimum, around 30 PSI
- Only a little cooling

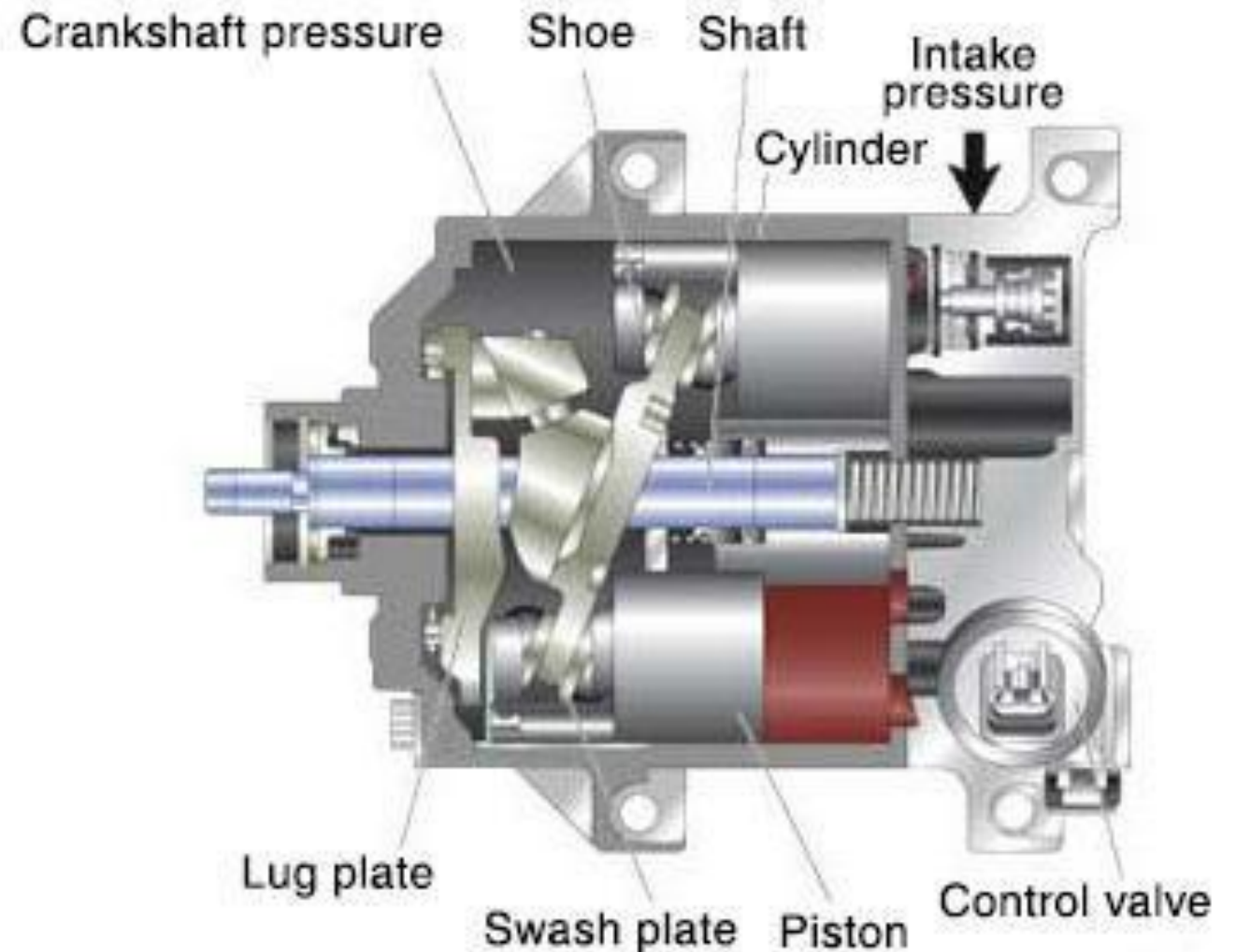


⚠ WARNING The variable compressor is very sensitive to debris.

Variable compressor

What to do

- This problem happens often with this type of compressor
- Make sure you have the right type and the right amount of oil
- Distribute the oil through the system
- If the wobble plate sticks, try the same procedure



The compressor with R1234yf



What changes on the compressor

- Slightly increased pumping capacity, e.g. VW from 137.2 cm³ to 140.3 cm³
- Computer-controlled variable compressor, used more and more, for a better pressure balance
- The clutch is eliminated in many cases, so the compressor then requires constant lubrication
- May be fitted with a flow sensor
- Lubricated with PAG oil for R1234yf

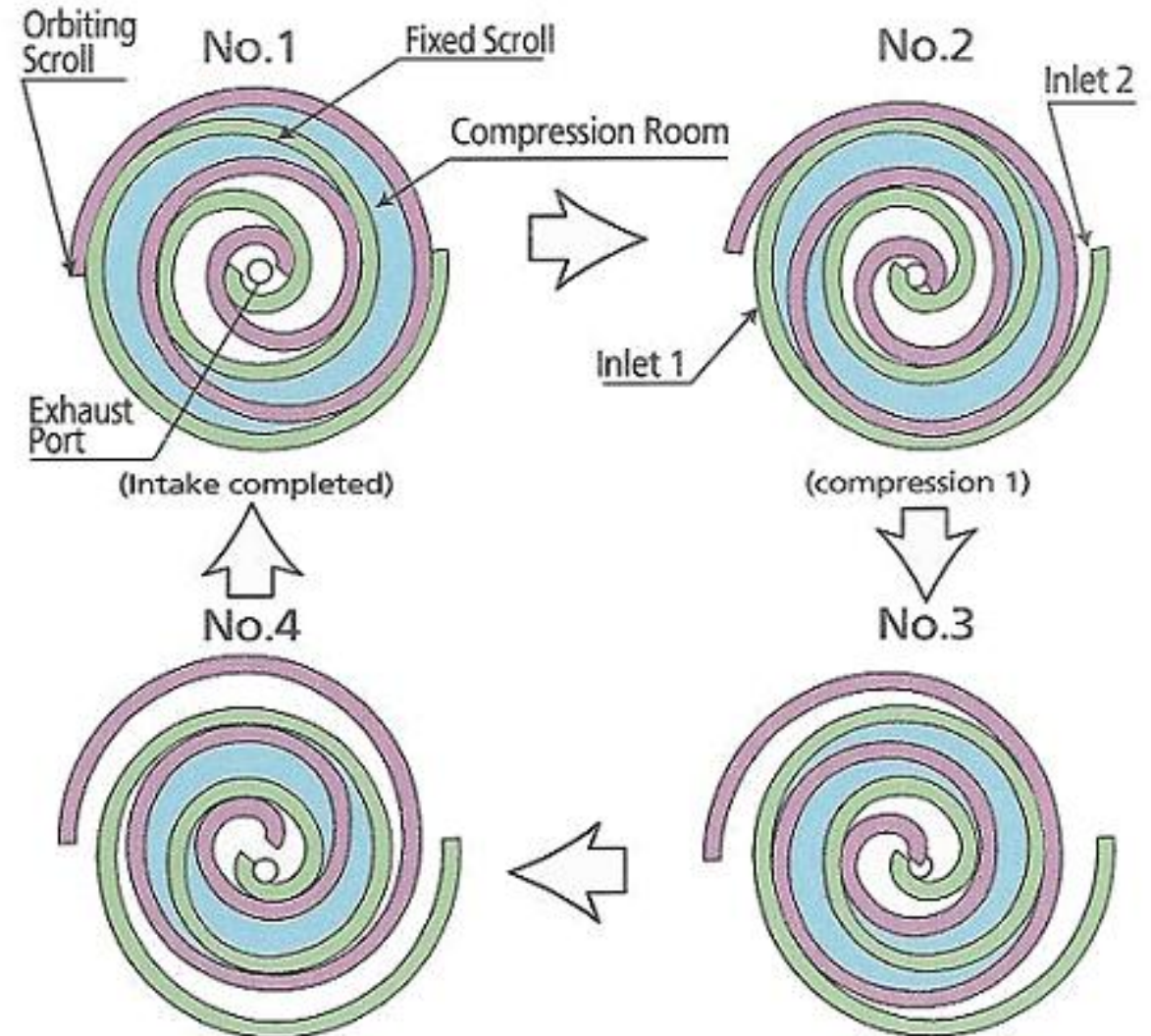
Scroll-type compressor

Scroll-type compressor

- These compressors are mounted in the lower part of the engine
- Problem: debris collects in the bottom of the system
- In the event of a failure, replace the condenser

Reference

- Honda CR-V 2002-2004: TSB 09-076



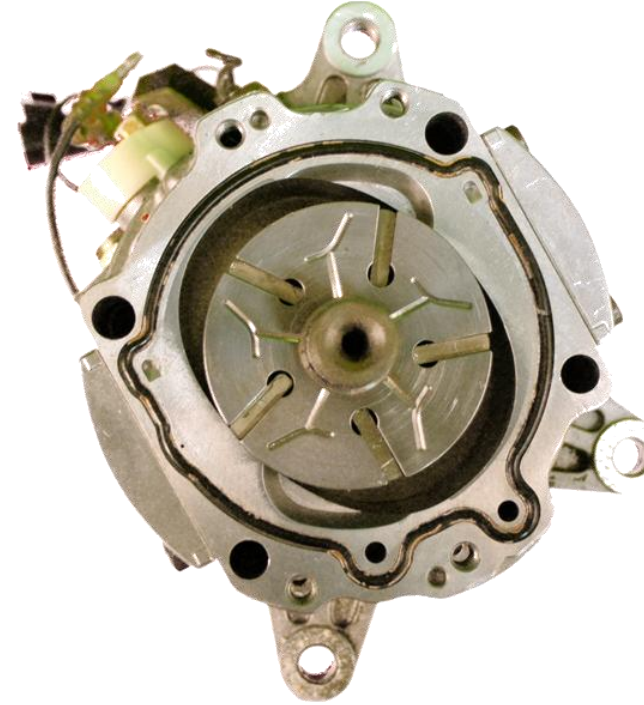
Rotary vane compressor

Symptom: poor performance

- Vehicles with a rotary vane type compressor
- Happens on the first use of the season, or right after the compressor is installed
- Pressures: low side high, high side low

Cause

- The compressor vanes stay stuck in their slots
- Infrequent use, wrong oil viscosity, or too much oil



Cause and correction: rotor vanes

Rotary vane compressor

Cause

- The compressor vanes stay stuck in their slots
- Infrequent use, wrong oil viscosity, or too much oil

Correction

- Recover half of the refrigerant charge
- Run the engine at 2 500 rpm
- Cycle the compressor repeatedly
- When the pressure comes up, recharge the system properly

- Generally installed on the compressor
- Detects the refrigerant flow
- Mounted in a 5 volt circuit
- Generates a reference voltage varying between .3 and 4.7 volts
- Voltage inversely proportional to the refrigerant flow (Toyota)
- Can prevent the A/C from engaging

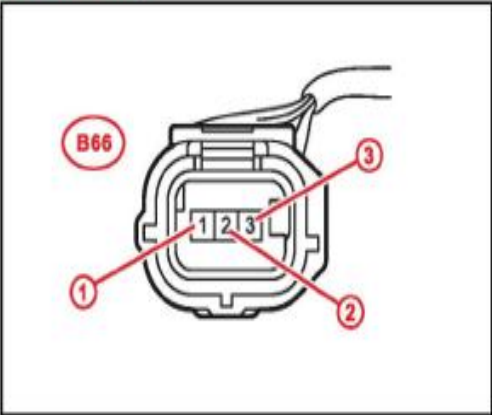


The flow sensor
(Toyota)



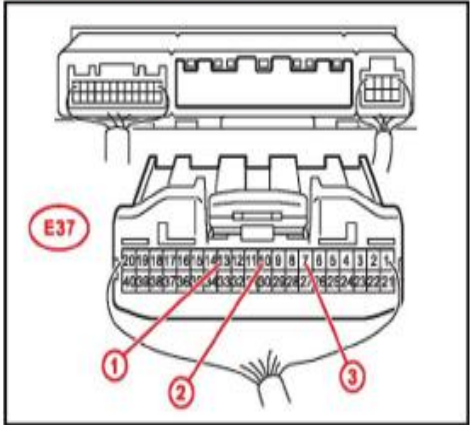
TESTER DISPLAY	MEASUREMENT ITEM/RANGE	NORMAL CONDITION
Flow Sensor	A/C Mass Flow Sensor / Min.: 0 V Max.: 5 V	A/C ON, Blower Switch HI: 0.7 to 3.8 V A/C OFF, Blower Switch OFF: 3.8 to 4.2 V

Figure 1. Front View of Wire Harness Connector
(to Flow Sensor)



1	S5FL
2	SGFL
3	QUFL

Figure 2. Rear View of Wire Harness Connector
(to Air Conditioning Amplifier)



1	SG-4
2	S5-1
3	FLOQ

Pressure Relief Valve



Performance checks

TEMPERATURE	HUMIDITY	GAGE PRESSURES		CENTER OUTLET TEMP	EVAP ASSEMBLY TEMP DISPLAYED ON TECH 1
		LOW	HIGH		
21°-27°C (70°-80°F)	LESS THAN 50%	110-165 kPa (16-24 psi)	793-1310 kPa (115-190 psi)	3°-9°C (38°-48°F)	1°-6°C (33°-43°F)
	HIGHER THAN 50%	145-200 kPa (21-29 psi)	793-1379 kPa (115-200 psi)	4°-13°C (40°-55°F)	2°-10°C (35°-50°F)
27°-33°C (80°-90°F)	LESS THAN 50%	131-207 kPa (19-30 psi)	965-1482 kPa (140-215 psi)	3°-13°C (38°-55°F)	1°-10°C (33°-50°F)
	HIGHER THAN 50%	165-269 kPa (24-39 psi)	1034-1620 kPa (150-235 psi)	7°-18°C (45°-65°F)	4°-16°C (40°-60°F)
33°-38°C (90°-100°F)	LESS THAN 40%	172-290 kPa (25-42 psi)	1138-1793 kPa (165-260 psi)	7°-17°C (45°-63°F)	4°-14°C (40°-58°F)
	HIGHER THAN 40%	234-296 kPa (34-43 psi)	1276-1862 kPa (185-270 psi)	13°-20°C (55°-68°F)	10°-18°C (50°-65°F)
38°-44°C (100°-110°F)	LESS THAN 20%	255-296 kPa (37-43 psi)	1448-2000 kPa (210-290 psi)	12°-18°C (53°-64°F)	9°-16°C (48°-61°F)
	HIGHER THAN 20%	262-352 kPa (38-51 psi)	1517-2137 kPa (220-310 psi)	14°-21°C (58°-70°F)	13°-18°C (55°-65°F)

Air Conditioning Performance Chart

A/C performance test

The reference measurement for the whole day

Conditions to respect

- Engine at operating temperature
- A/C system at minimum temperature
- Engine at 1 500 rpm
- Doors open
- Thermometer in the centre right vent
- Wait 3 minutes for it to stabilize

The magic number: DELTA 30 °F

- Ambient temperature, measured 12 to 18 inches in front of the condenser
- MINUS the centre vent temperature

Low-side pressure analysis

Low side

Cycling system

- Vehicle under performance test
- Average the pressures at compressor cut-out and cut-in
- E.g.: clutch ON 41 PSI, clutch OFF 19 PSI $\rightarrow (41 + 19) \div 2 = 30$ PSI

Variable system

- Vehicle under performance test
- Read the low side at idle, then at 2 000 rpm
- E.g.: idle 28 PSI, revved 32 PSI $\rightarrow (28 + 32) \div 2 = 30$ PSI

 **KEY POINT** Specification: between 28 and 32 PSI for both systems.

High-side pressure analysis

High side

Two criteria determine the expected high side

- Ambient temperature (°F)
- Humidity level (%)

Formula based on the humidity level

- Humidity between 0 % and 33 %: $\text{ambient} \times 2 + 10 \%$
- Humidity between 33 % and 66 %: $\text{ambient} \times 2.5 + 15 \%$
- Humidity between 66 % and 100 %: $\text{ambient} \times 3 + 20 \%$

Example

- $85 \text{ }^{\circ}\text{F} \times 2.5 + 15 \% = 245 \text{ PSI}$

Performance test

Condenser heat rejection

What we are checking

- How efficiently the condenser rejects heat

Method, vehicle under A/C performance test

- Convert the high side into a temperature
- Measure the ambient temperature
- Take the difference between the two

Example

- High side = 175 PSI, conversion: 123 °F
- Ambient 75 °F → $123 - 75 = 48$ °F

Specification

- Less than 60 °F

Pressure ratio

Measuring how hard the compressor is working

What the ratio tells you

- The effort the compressor puts in for its performance to be optimal

Method, vehicle under A/C performance test

- Take the low-side and high-side values
- Add atmospheric pressure (14.8 PSI) to both values
- Divide the high side by the low side

Example

- High side: $195 \text{ PSI} + 14.8 \text{ PSI} = 209.8 \text{ PSI}$
- Low side: $32 \text{ PSI} + 14.8 \text{ PSI} = 46.8 \text{ PSI}$
- $209.8 \div 46.8 = 4.482:1$

Specification

- Maximum 7:1

 **KEY POINT** The ratio is calculated in absolute pressure. If you forget to add the 14.8 PSI of atmospheric pressure to both readings, the result is skewed and looks higher than it really is.

A/C compressor

What the compressor temperature is telling you

Compressor temperatures

- In normal operation: between 95 and 140 °F
- At 200 °F: the compressor is at its maximum
- At 225 °F: the oil starts to burn
- At 250 °F: the compressor blows up

Cycling rate

- This is the number of ONs and OFFs the compressor makes in one minute

⚠ WARNING Between 200 °F and 250 °F, there are only 50 degrees between « at its maximum » and « it blows up ». Compressor temperature is a signal to take seriously right away.

Cycling rate

The number of ONs and OFFs per minute

How to measure it

- Vehicle under A/C performance test
- Count the number of cycles with the doors open
- Then the number of cycles with the doors closed

Specification

- Maximum 4 cycles per minute

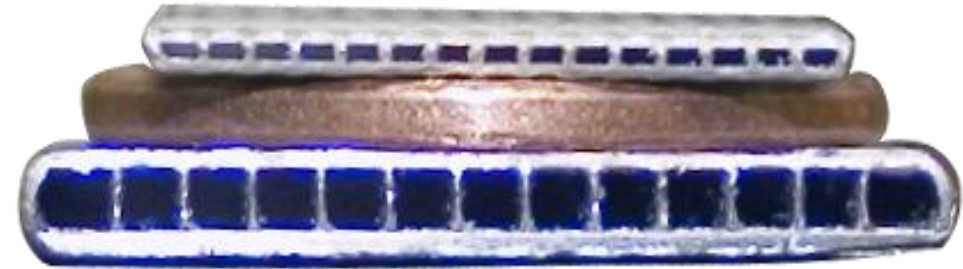
Causes of a cycling rate that is too high

- Incorrect system charge
- Internal restriction
- Electrical problem
- Clutch electromagnet

After a compressor failure

The rule

- Every inch of refrigerant passage should be new or cleaned



What cannot be flushed

- Flat-tube and multi-pass condensers



What can be flushed

- The evaporator, the hoses and the lines without mufflers

Flushing essentials

► What a good solvent must be

- Good quality
- Evaporates quickly, leaves no residue
- Compatible
- Safe



Here is a good tool

Procedure

- Pressurize the system with the flush solvent
- Let the product soak for 10 minutes
- Flush with nitrogen at 50 PSI
- Until it is dry



Condensers

How to check a condenser

- Measure the inlet temperature against the outlet temperature

Specification

- The difference must be between 20 °F and 50 °F

Condenser inlet (vapour)



Condenser outlet (liquid)



Performance test (condenser)

Condenser subcooling

What we measure: subcooling (latent heat)

- Convert the high-side pressure into a temperature
- Measure the condenser outlet temperature
- Take the difference between the two

Range: 10 °F to 25 °F, perfect at 16 °F

- Below 10 °F: not enough refrigerant
- Above 25 °F: too much refrigerant

Interpreting the results

Condenser subcooling

If below 10 °F

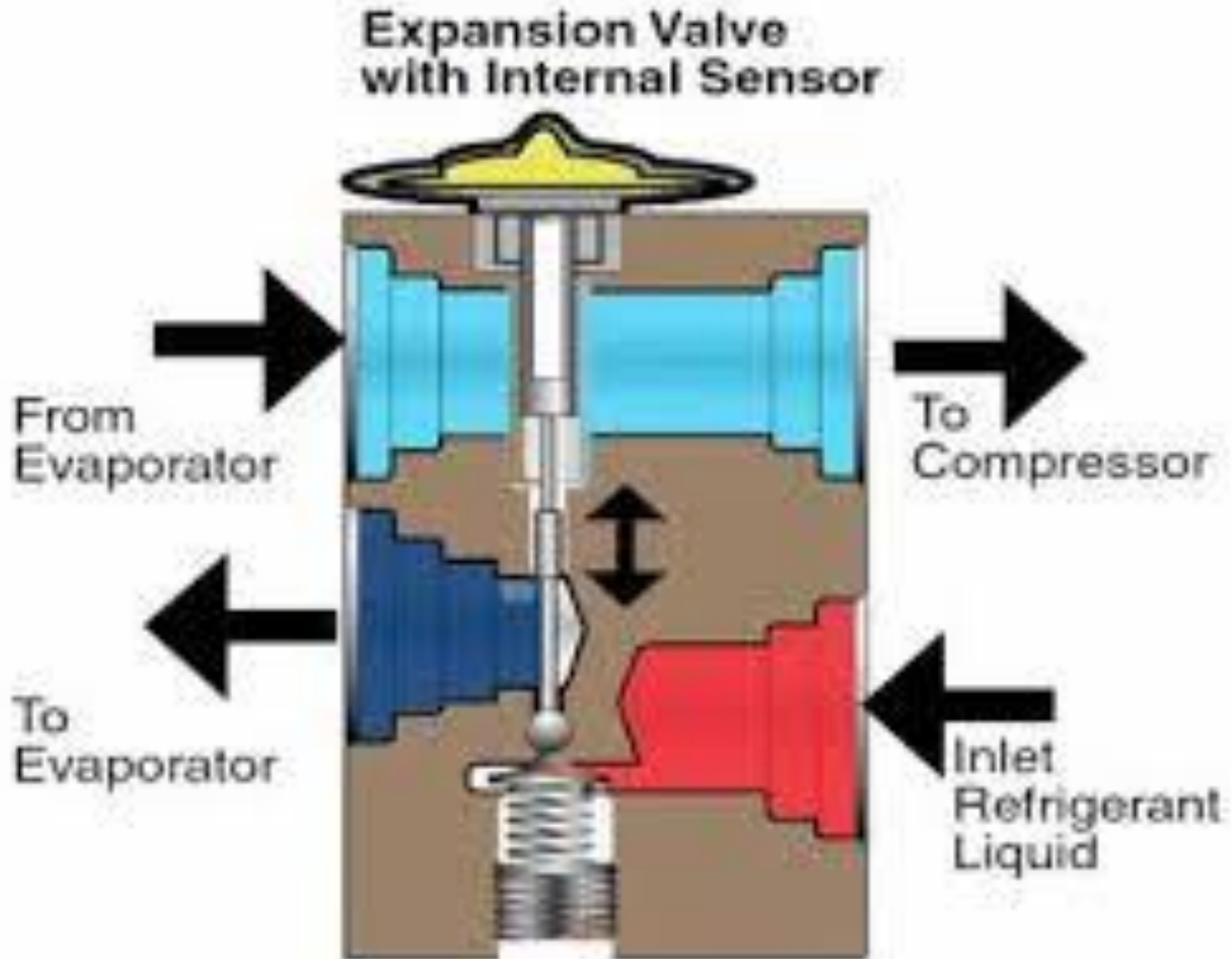
- System undercharged
- Bad condenser
- Poor air transition
- Contaminated refrigerant
- Faulty compressor

If above 25 °F

- System overcharged
- Blocked condenser
- Blocked expansion valve

 **KEY POINT** Subcooling range: 10 °F to 25 °F, perfect at 16 °F. Below that, not enough refrigerant; above that, too much refrigerant.

The H-block expansion valve



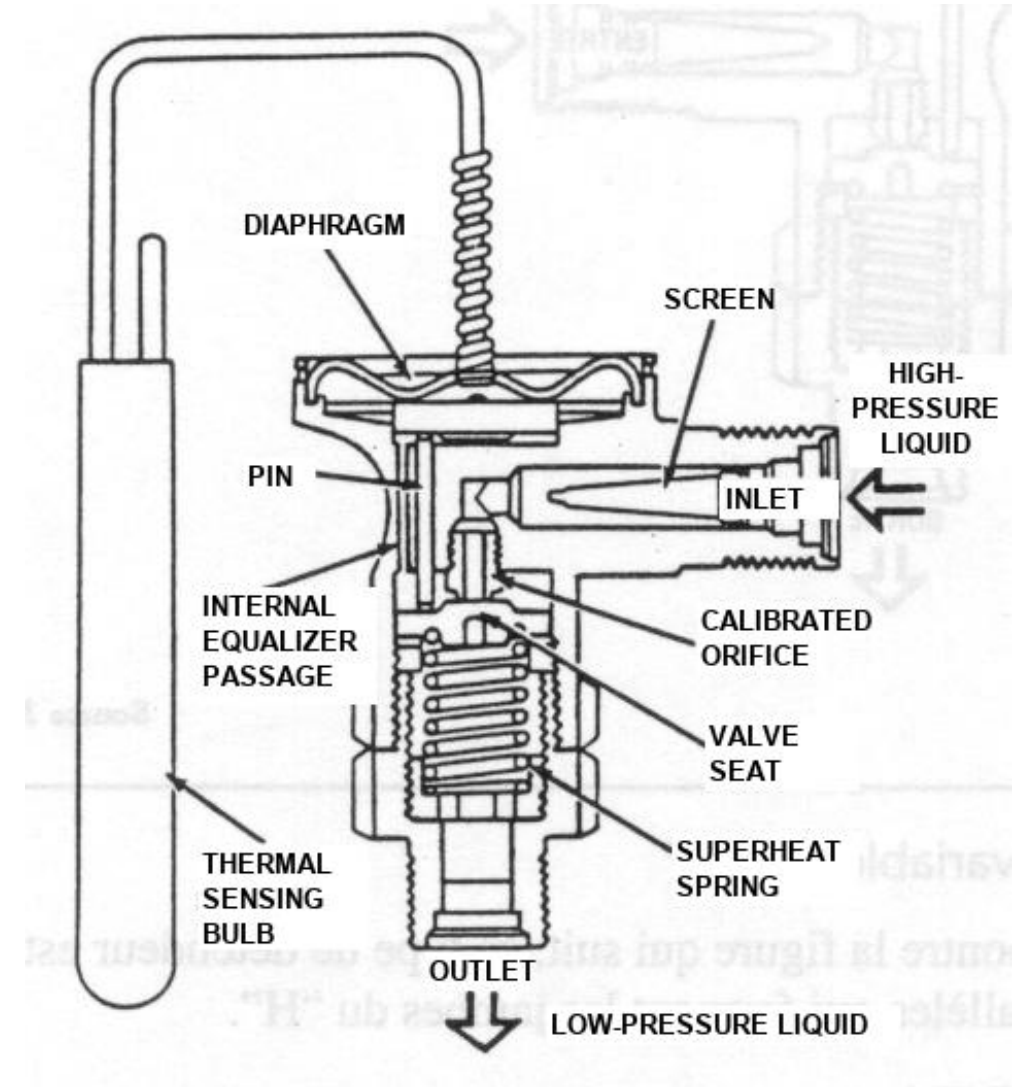
Internally compensated variable expansion valve

How the orifice varies

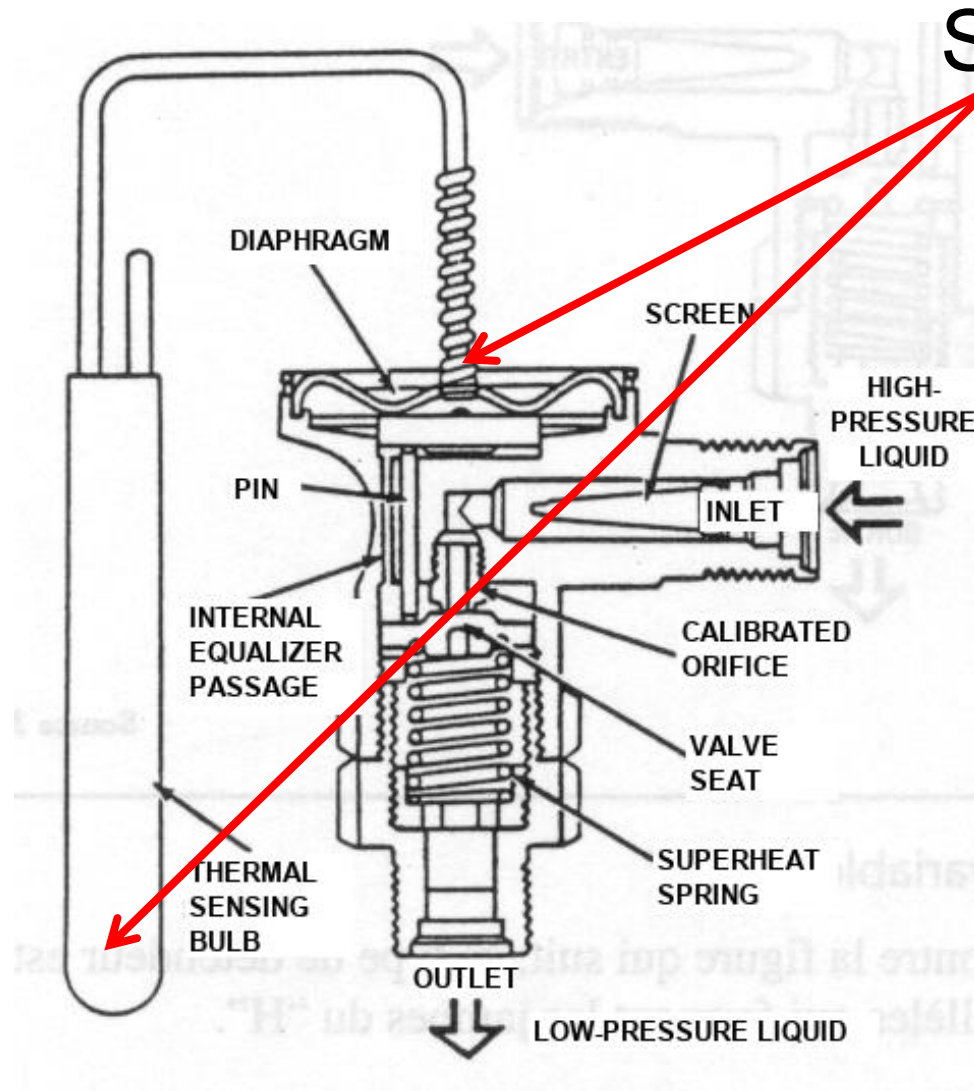
- A diaphragm closes under the pressure coming from the evaporator (lower section)
- Or under the action of a return spring (upper section)
- A very fine mesh screen filters impurities at the high-pressure inlet

The role of the external bulb

- It connects to the body of the valve through a thin capillary tube
- Attached to the low-pressure line, near the evaporator outlet
- Filled with a gas that has a high expansion coefficient
- It expands or contracts with the slightest change in temperature, which stabilizes the valve



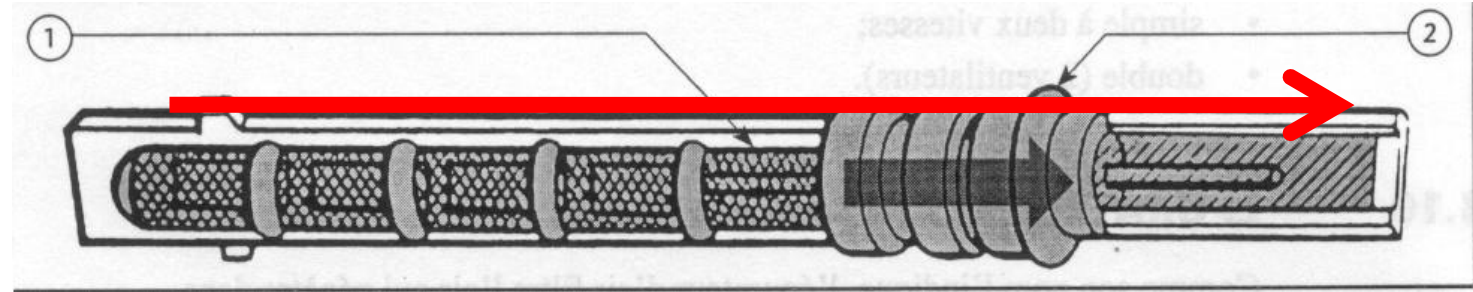
To check the expansion valve



Spot to warm up in order to
raise the temperature.
(HEATER GUN)

**IF THERE IS A
VARIATION, THE
EXPANSION
VALVE IS GOOD.**

Fixed orifice tube



What it is

- A small plastic tube with a calibrated orifice
- A strainer (filter) at each of its ends

Its real role

- It does not control the amount of refrigerant, it acts as a restrictor
- It raises the refrigerant pressure at the condenser outlet
- Combined with the drop in flow, the pressure drop produces a low-pressure liquid that enters the evaporator

Black = 0.072"

Blue = 0.067"

Brown = 0.047"

Red = 0.062"

Orange = 0.057"

Green = 0.052"

Receiver-drier

► What it is

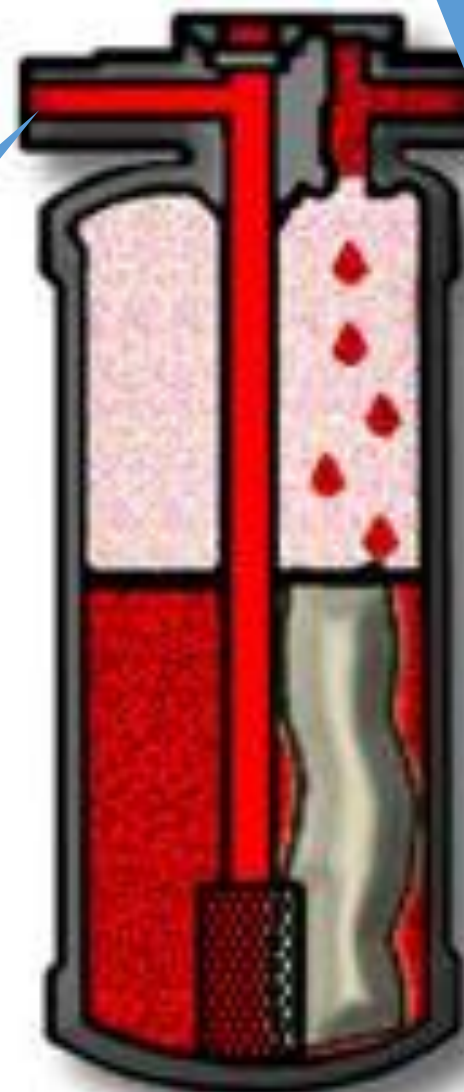
- A reservoir that filters the refrigerant coming from the condenser
- Upper section: a desiccant
- Lower section: a screen that holds back the impurities that could damage the compressor

► Good to know

- Sometimes fitted with a sight glass that lets you judge the refrigerant level
- Service life: 3 years, or if the circuit has been left open for 24 to 48 hours

Outlet

Inlet



Critical solubility threshold

What the critical threshold tells you

- The temperature change at the point where the oil separates from the refrigerant

What causes opposite conditions

- Low refrigerant level
- Internal restrictions
- Poor airflow through the condenser
- Inadequate cooling system

Normal condition

Oil circulation 10-25%



Overheating conditions

oil circulation 2-4%

Oil



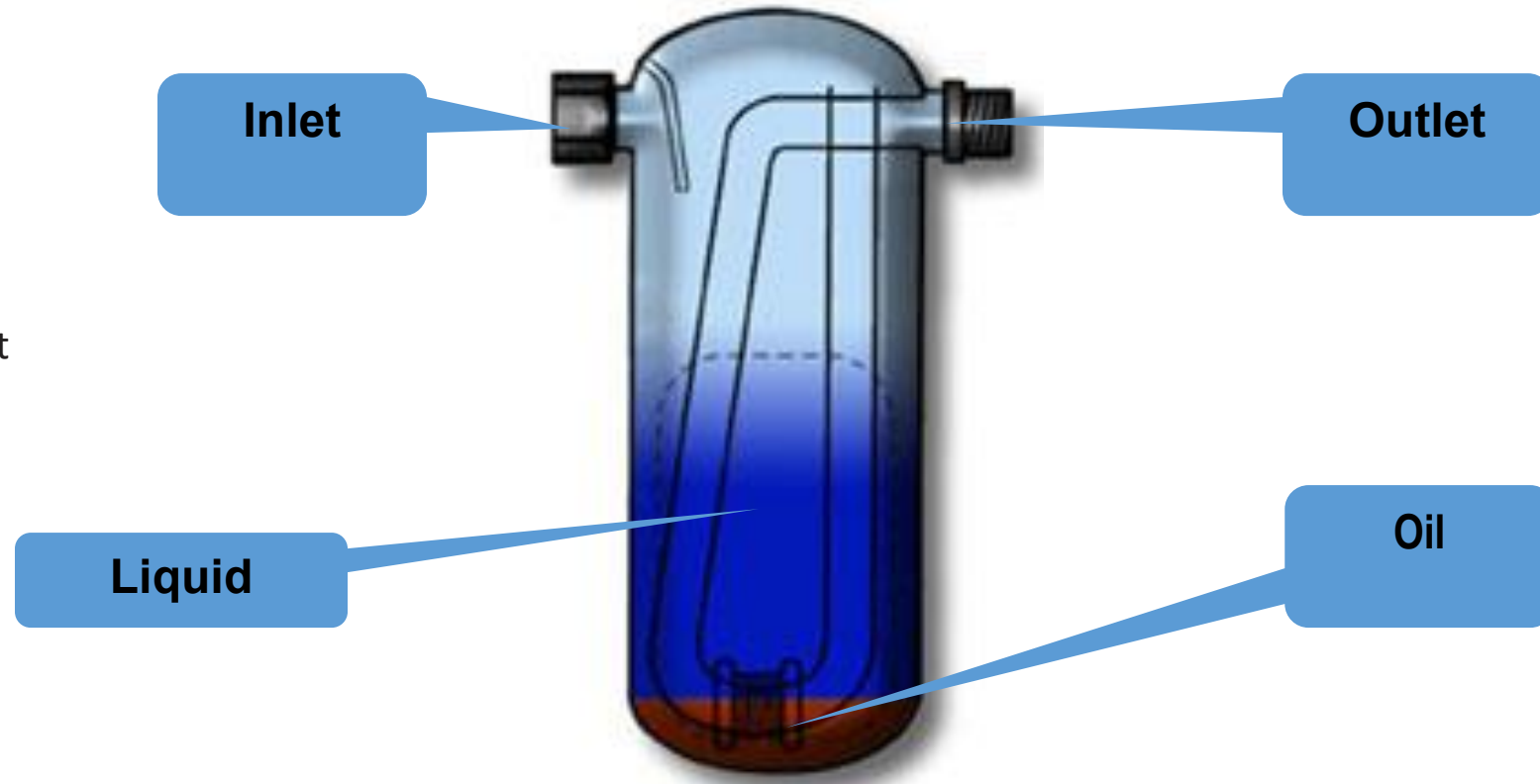
Accumulator-drier

► What it is

- Connected to the evaporator outlet tube
- It separates the refrigerant from the oil and the moisture, so that only pure refrigerant goes to the compressor
- It contains a desiccant (silica gel for example) that holds back water vapour
- A bleed orifice at the bottom returns the oil that settles there

► Can a drier be tested? YES

- With the vacuum produced by the pump



► **Measuring the vacuum**

- Use a micron gauge
- Target: between 500 and 900 microns



The evaporator... a new standard - R-1234yf

Why the evaporator changes with R-1234yf

- The changes are safety related
- R1234yf is mildly flammable and the evaporator sits inside the cabin
- Hence thicker walls

What the standard requires

- OEM and aftermarket manufacturers must design the evaporator to meet the safety standards
- Standard SAE J2842 had to be brought into force, and manufacturers must comply with it



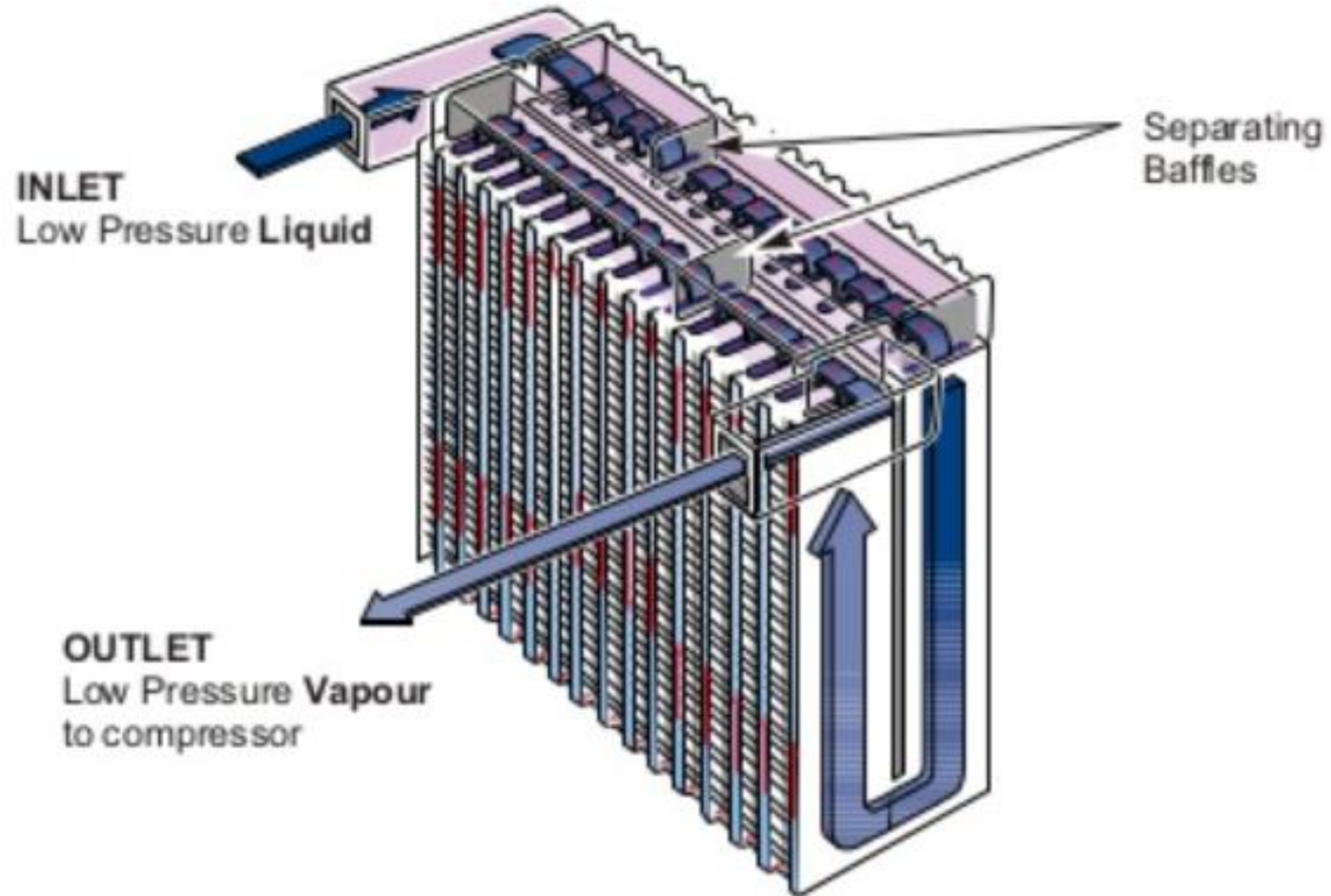
Evaporator (with orifice tube)

Measuring superheat (sensible heat)

- Vehicle under performance test
- Temperature between the evaporator inlet and outlet

Specification

- Outlet – inlet = -5°F to $+5^{\circ}\text{F}$



Interpreting the results

Superheat, orifice tube system

Below -5°F

- Too much refrigerant
- Wrong orifice tube
- Poorly sealed orifice tube
- Insufficient airflow at the condenser and the evaporator

Above $+5^{\circ}\text{F}$

- Not enough refrigerant
- Air present in the system
- Orifice tube restriction
- Restriction between the condenser and the expansion valve

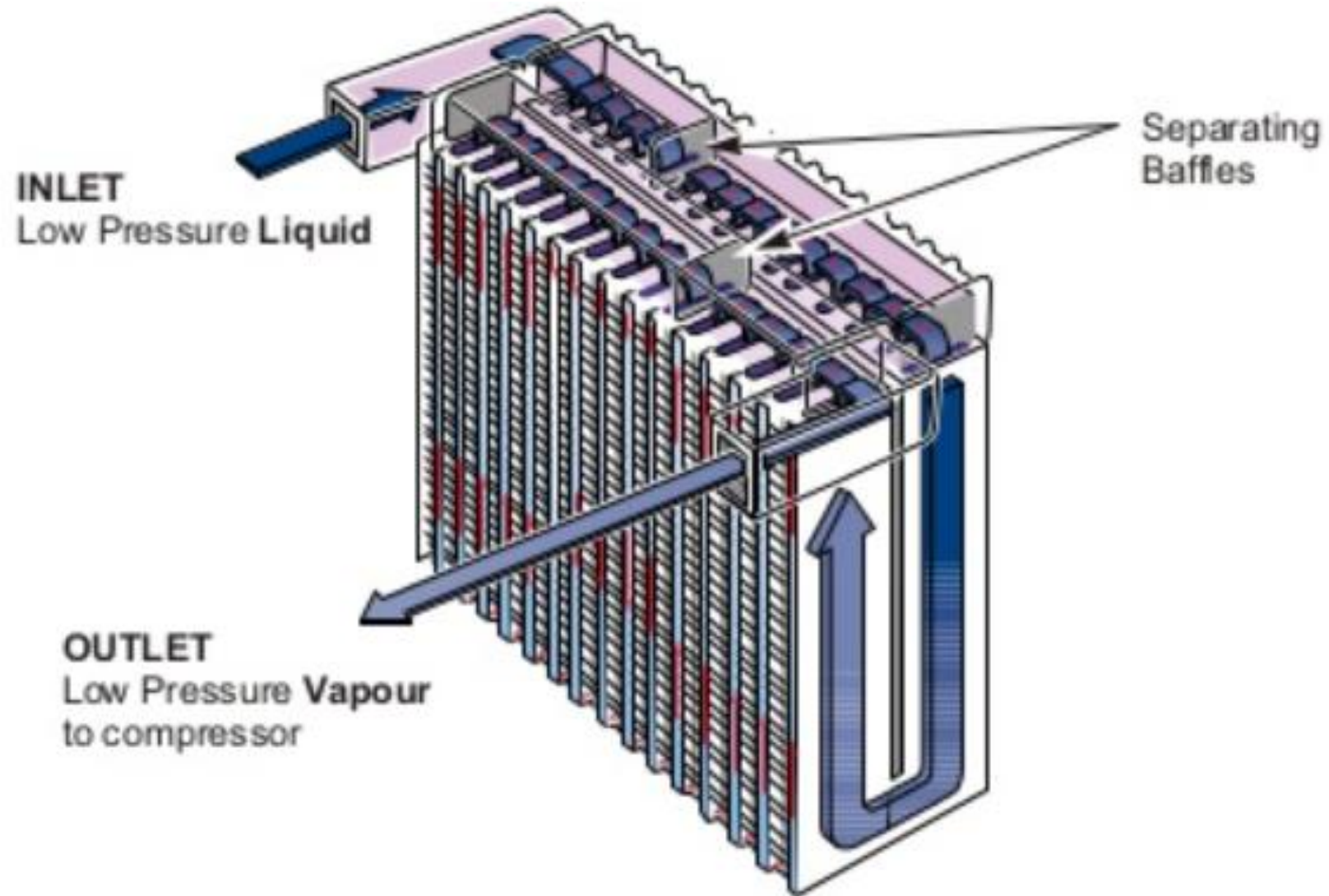
Evaporator (with expansion valve)

Measuring superheat (latent heat)

- Vehicle under performance test
- Measure the evaporator outlet temperature
- Convert the low side into a temperature

Specification

- Evaporator outlet – converted low side = 4 °F to 16 °F



Interpreting the results

Superheat, expansion valve system

If the difference is below 4 °F

- System overcharged
- Expansion valve metering too much
- Insufficient airflow at the condenser and the evaporator

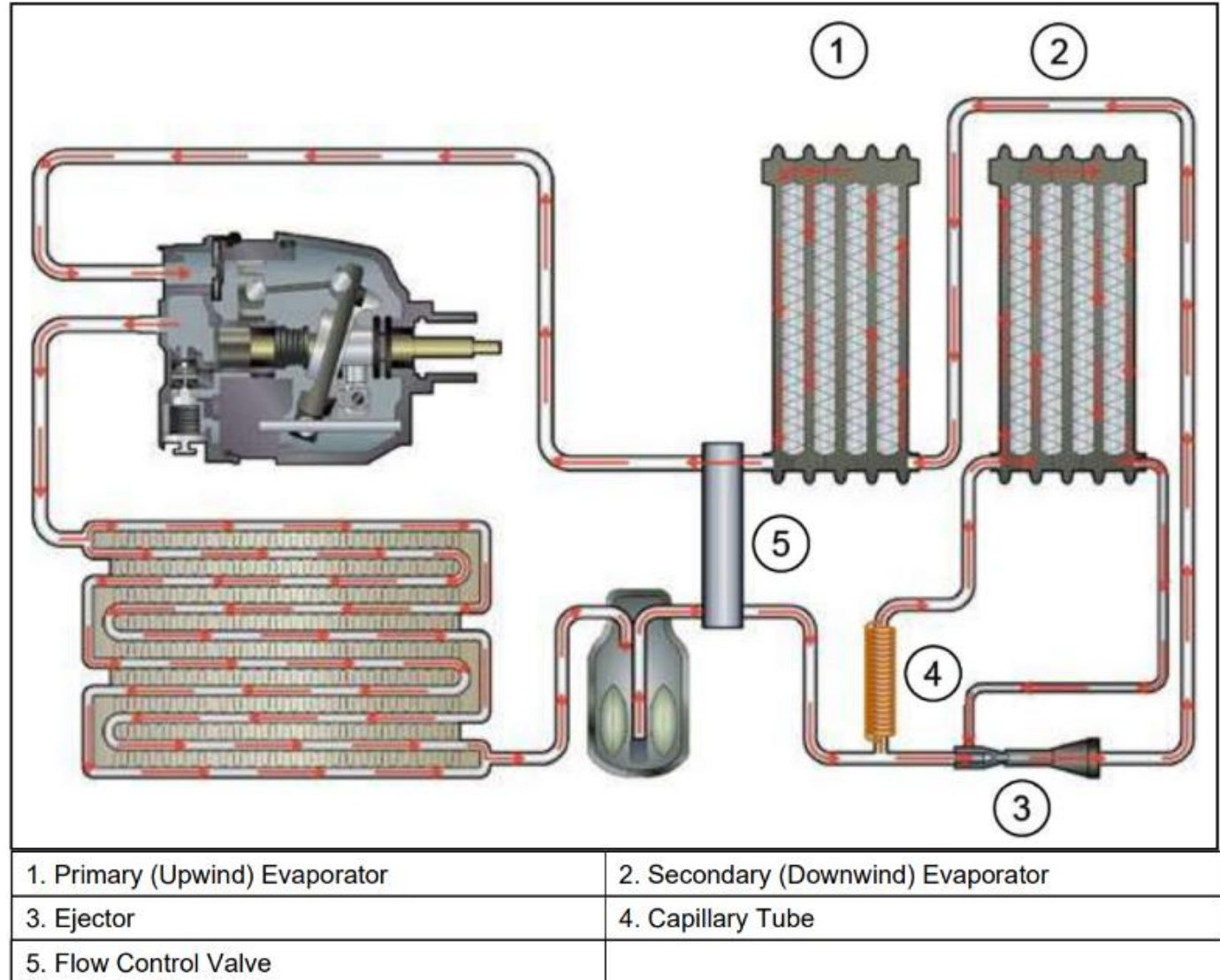
If the difference is above 16 °F

- Expansion valve stuck closed
- System low on charge
- Restriction between the condenser and the expansion valve

Evaporator (with ejector cycle)

Application

– Toyota Prius 2010



Lubrication

To check before anything else

- The TSB, for a possible change in capacity
- The system capacity and the oil type

Good to know about PAG

- It is hygroscopic, so buy it in a small container

How to split the oil when adding it

- Orifice tube system: $\frac{1}{2}$ in the compressor, $\frac{1}{2}$ in the accumulator
- Expansion valve (TXV) system: $\frac{1}{2}$ in the compressor, $\frac{1}{2}$ in the evaporator
- Turn the compressor by hand, 15 turns



Characteristics of PAG oils

high miscibility with the refrigerant

good lubricating properties

contain no acids

very hygroscopic (absorbs water)

not miscible with other oils

Characteristics of the lubricants

R134a

- PAG (polyalkylene glycol), viscosities 46, 100, 125, 150
- POE (polyol esters), viscosity 100

Solubility

- This is the ability of the oil to dissolve in the refrigerant



Use the right type of oil

Oil for hybrid A/C (POE)

Why a special oil

- Some hybrid vehicles have an electric A/C compressor
- PAG and ESTER conduct electricity, so they must not be used

Not to be forgotten

- Even the dye must be approved for hybrid vehicles



Oil for hybrid A/C (PVE)

What PVE (polyvinyl ether) is

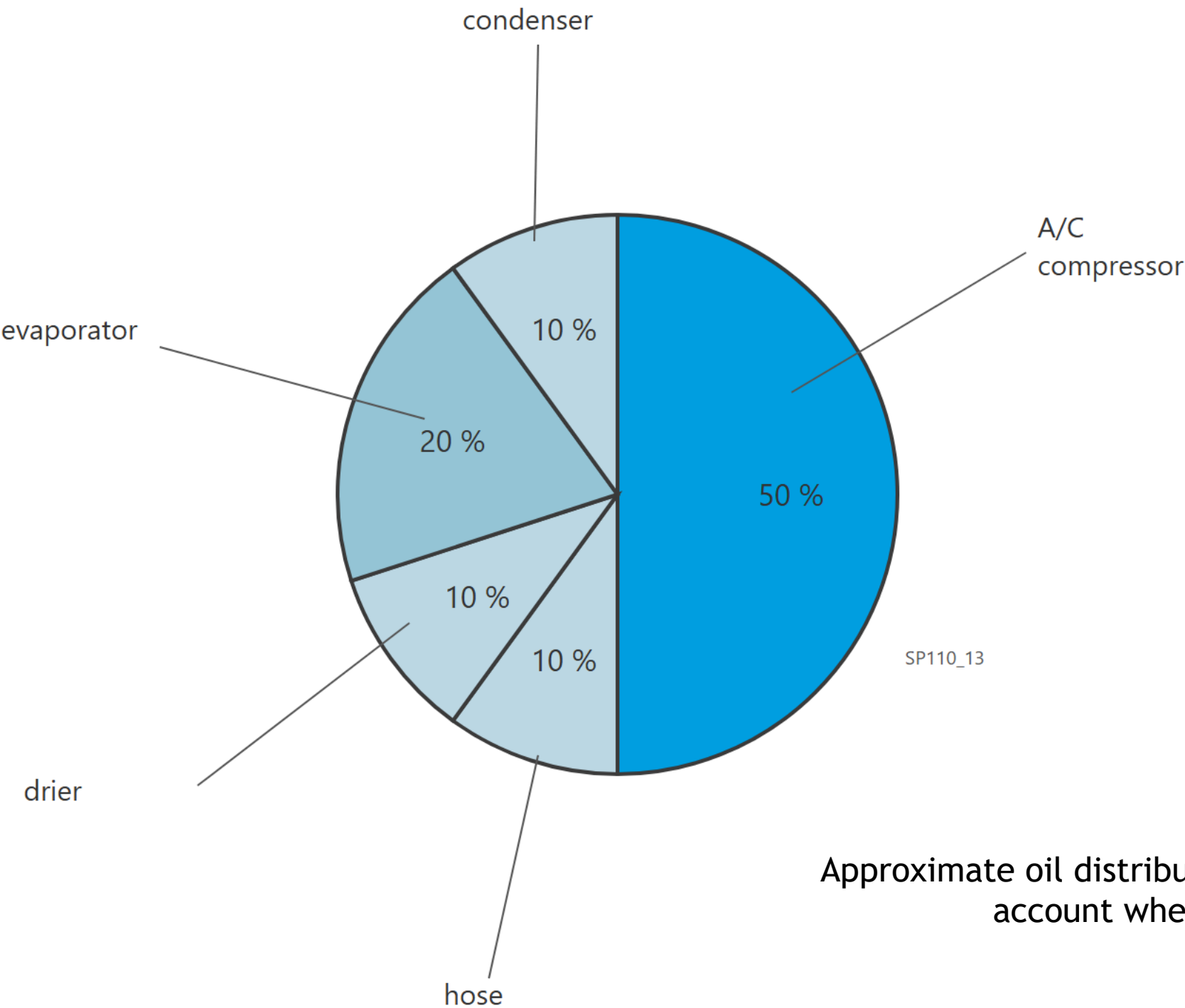
- Used in some HEVs as an alternative to POE oil
- Better electrical resistance
- Less hygroscopic and resistant to hydrolysis, so it does not react with water
- Compatible with R-134a and R-1234yf systems
- ISO 68 type viscosity

Where you find it

- Volvo EX30 2025, Polestar 4 2026
- Korean manufacturers, 2024-2025 and newer

⚠ WARNING This only concerns certain models. You have to confirm with the vehicle shop manual before choosing the oil.

How the oil is distributed in the air conditioning system



SP110_13

Approximate oil distribution the technician should take into account when replacing a component

Is the system low on oil?

Diagnosing a lack of oil

The method

- Measure the compressor temperature
- Measure the A/C outlet temperature

The correction

- Add the proper oil in small amounts, maximum ½ oz at a time
- Until the compressor temperature stabilizes
- Without affecting the A/C outlet temperature

Checking static pressure

What a correct static pressure proves, and what it does not

How to read a static pressure

- After 10 minutes at rest
- The low-side and high-side gauges must be equal
- And must match the ambient temperature

⚠ WARNING A correct static pressure does not tell you the system is full, only that it holds 30 % or more of its charge.

Sealers

– Your call.

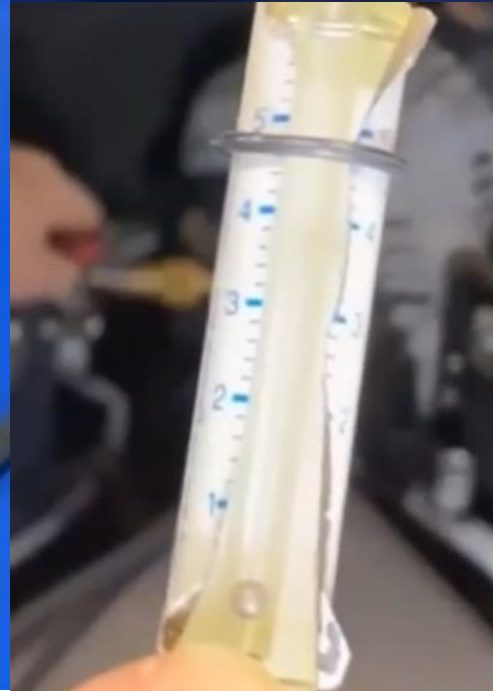


Procedure with the leak sealer unit



► Procedure

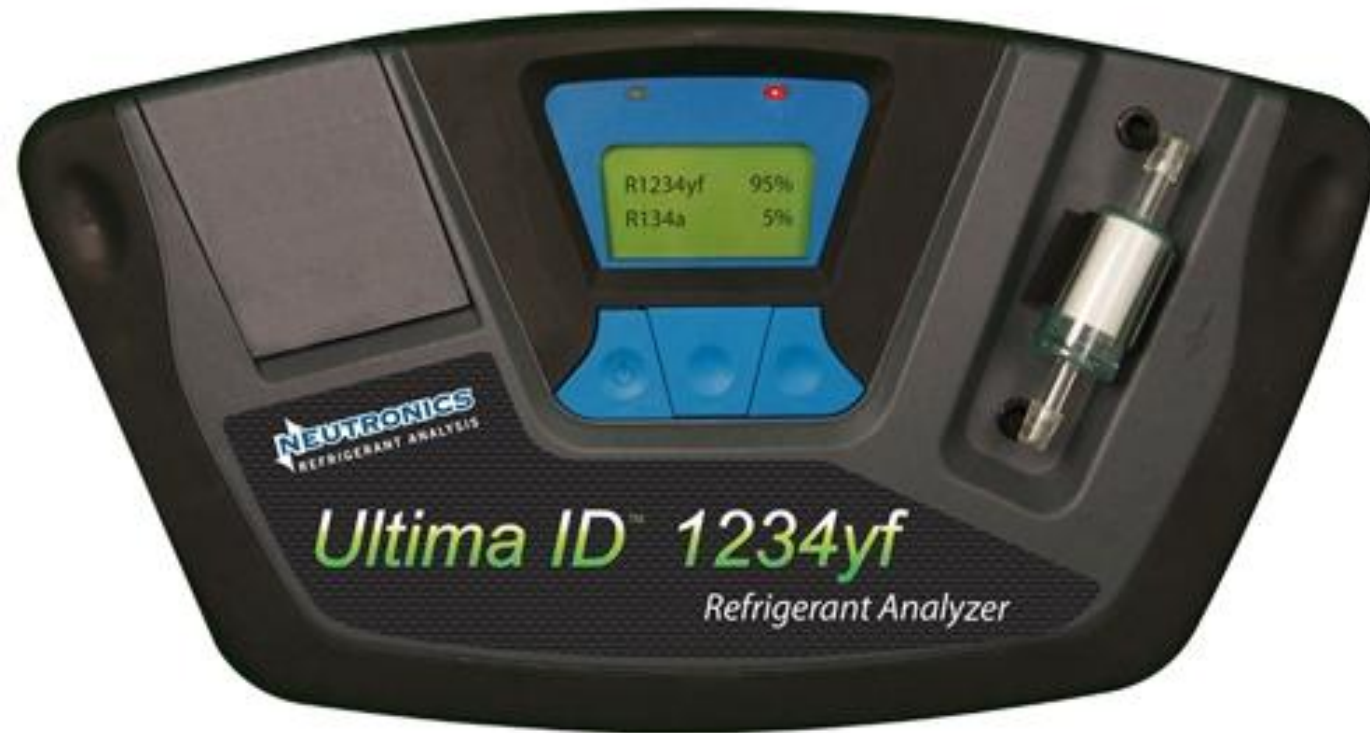
- Run the vehicle at normal temperature, A/C on
 - Shut the engine off and let it settle for 3 to 5 minutes
 - Static pressure minimum 60 PSI
 - Wet the sleeve on both sides and shake
 - Connect with the tool
 - Let it settle for 5 minutes
- MAXIMUM



If there is leak sealer in the system



Refrigerant analysis



Analysis thresholds

- R134a or R1234yf: 98 % minimum
- R-22: 2 % maximum
- HC: 2 % maximum
- Air: 3 % maximum

Good to know

- An additional calculation is done by the unit

The R1234yf refrigerant identifier

What it is for

- Analysing the contents before draining a system

Standard

- Must meet standard SAE J2912

Rule for reading it

- R134a detected in a system that normally uses R1234yf must be treated as a contaminant



If the refrigerant is contaminated

What must be done

- Empty the whole system and do an A/C flush
- Recovery is done with another machine, INDEPENDENT from the one used for R-134a or R-1234yf



Doing a recovery on your system

Refrigerant recovery

Not to be forgotten

- Work with the Government form

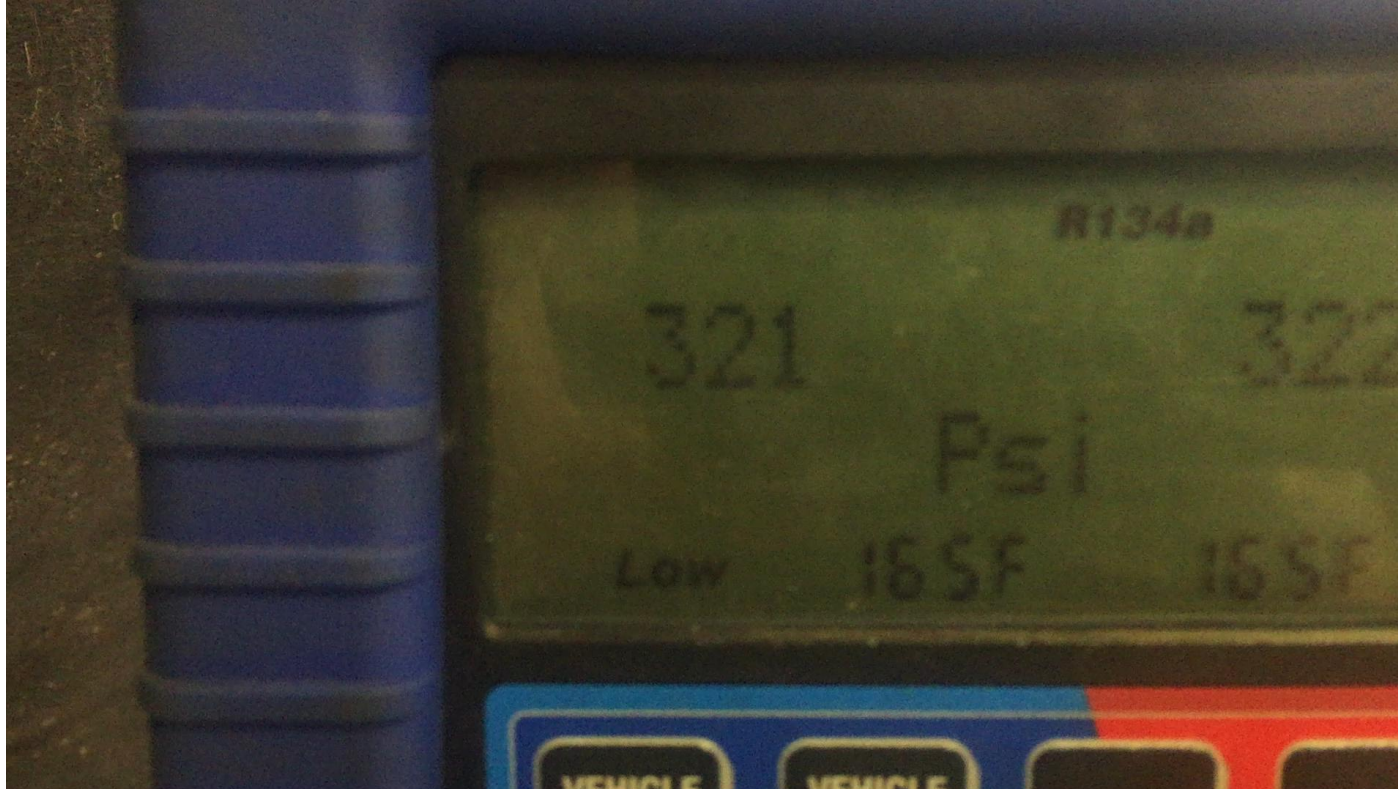
If refrigerant is missing, there is a leak

- Do a PRESSURE leak test with nitrogen

Recovery tip

- Warm up the drier to evacuate faster

A drop of 1 PSI in 20 minutes is tolerated



Using a digital gauge is far more precise because it tells you which side is leaking

**Suggested
product = BIG
BLUE**



Efficiency test

What it is

- A test often available with the diagnostic monitor on modern air conditioning systems

What it tells you

- Whether the air conditioning system is running with the recommended refrigerant charge





<Usage>

Use to judge whether the filling of the refrigerant gas is necessary.

<Introduction>

Confirm how much refrigerant gas is left and decide whether refrigerant gas should be filling.

OK



12:21



Refrigerant gas volume check

VCI

13.21V

The Refrigerant Gas Volume Check will be performed.

Please confirm the following conditions.

- IG ON
- Engine is ON
- Air Conditioner ON
- Blower Motor High
- Ambient temperature between 0.0°C (32.0°F) and 49.0°C (120.2°F)

Press Next to perform the operation.

Next

Exit



12:21



Refrigerant gas volume check

VCI  13.26V

The Refrigerant Gas Volume Check has completed.
the gas volume is normal.

ESC



12:22

Pulling a vacuum on the system

– Why we pull a vacuum

VACUUM *			Water boiling point (°C)
inches of Hg	mm of Hg	kilopascals	
0	760	101.3	100
5	633	84.3	95
10	506	67.4	89
15	379	50.5	81
20	252	33.6	72
25	125	16.7	56
29	23.4	3.1	25
29.5	10.7	1.4	12

* this really is vacuum, that is, pressures below normal atmospheric pressure

Temperature in °F	Inches of vacuum	Micron*	Pressure in lb/in ²
212°	0.00	759 968	14 696
205°	5.00	5354 000	12 279
194°	9.81	525 256	10 162
176°	16.02	355 092	6 866
158°	20.80	233 680	4 519
140°	24.12	149 352	2 888
122°	26.36	92 456	1 788
10°4	27.83	55 118	1 066
86°	28.75	31 750	0.614
80°	29.00	25 400	0.491
76°	29.10	22 860	0.442
72°	29.20	20 320	0.393
69°	29.30	17 780	0.344
64°	29.40	15 240	0.295
59°	29.50	12 700	0.246
53°	29.60	10 160	0.196
45°	29.70	7 620	0.147
32°	29.82	4 572	0.088
21°	29.90	2 540	0.049
6°	29.95	1 270	0.0245
-24°	29.99	254	0.0049
-35°	29.995	127	0.00245

* Surplus pressure in the system, in microns

1,000 inch = 25 400 microns = 2.540 cm = 25.40 mm

0.100 inch = 2 540 microns = 0.254 cm = 2.54 mm

0.039 inch = 1000 microns = 0.100 cm = 1.00 mm

This scale gives you the pump running time needed to remove the moisture completely.

Charging your system

The amount in the hoses is part of the charge

Charging rules

- Always add the required amount
- Do not forget to add the refrigerant held in the hoses (J2210)
- 2 oz (0.25 lb or 0.060 kg) per 6 to 8 foot hose

J2788 machines

- They perform an equalize hoses

Charging the system under the R-1234yf standard

Step 1: pressure test

- The system is first charged to 15 % of its capacity

Step 2: leak search

- Leak detector placed on the front floor, driver side
- Blower ON at speed 1, air directed to the floor
- Check whether there is a refrigerant leak
- Same check at the rear evaporator if there is one

Step 3: final charge

- If no leak is detected, go ahead with the final charge of the system

Electronic leak detector

What the standard requires

- A leak detector that meets standard SAE J2913
- Required to repair R1234yf air conditioning systems

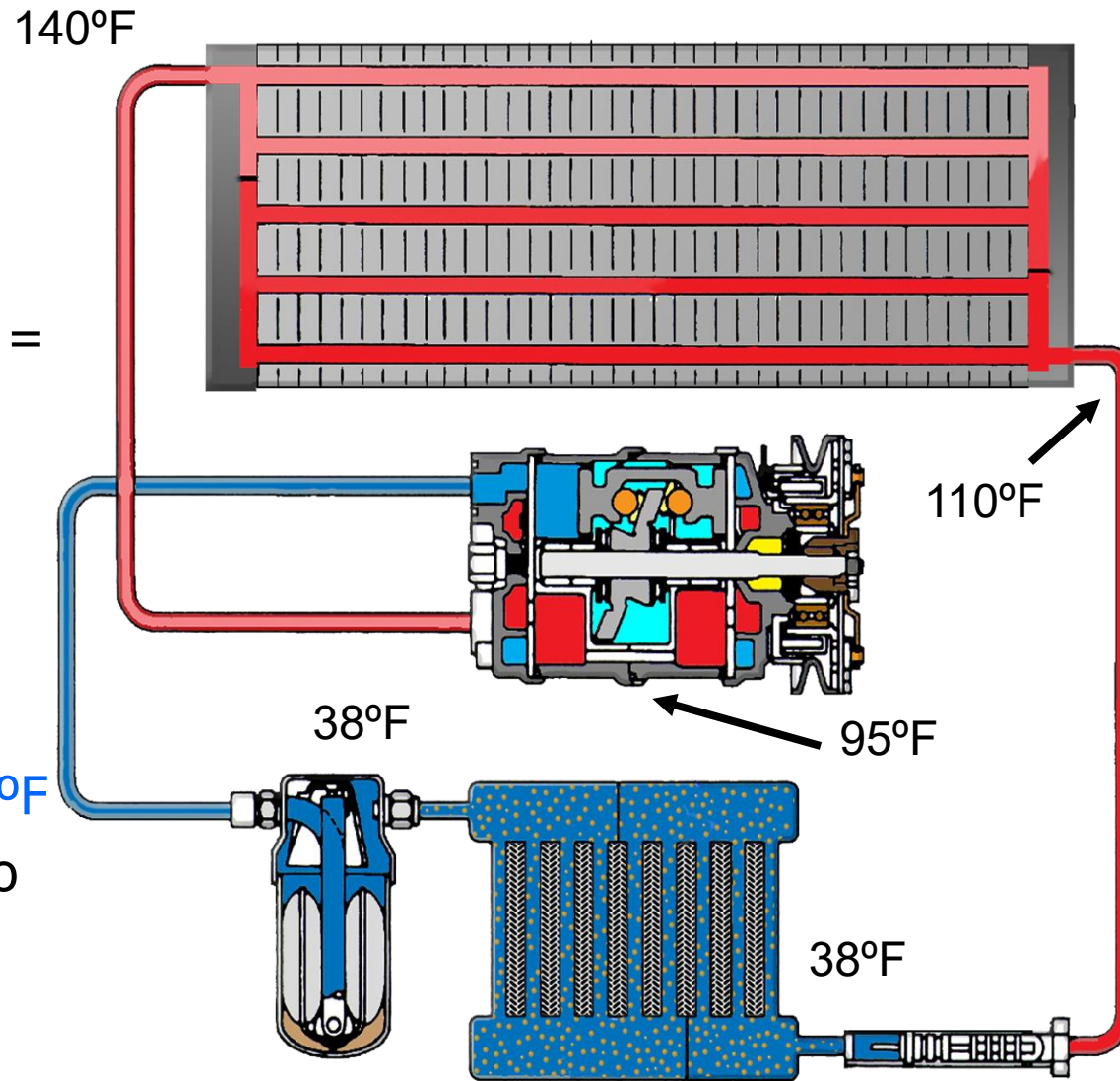
How it is used

- It is required in order to check the system evaporator



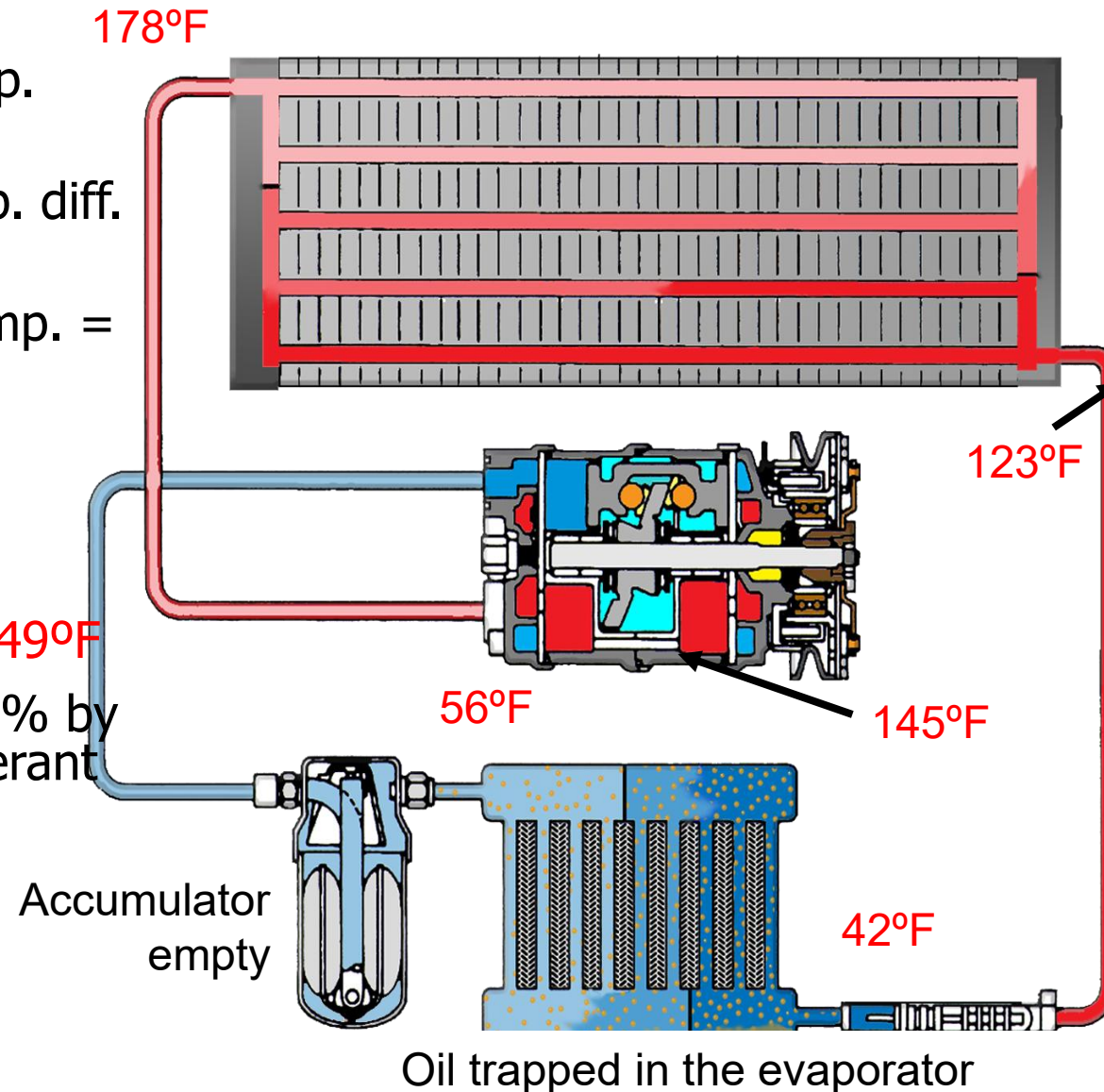
Normal charge / adequate oil level

- Condenser temperature drop = 30°F
- Evaporator temperature drop = 0°F
- Compressor temperature 95°F
- Ambient air temperature 88°F
- Air vent (duct) 46°F
- Oil circulation 10 to 25% by weight of refrigerant



Low charge / reduced oil circulation

- Condenser temp. drop = **55°F**
- Evaporator temp. diff. = **+14°F**
- Compressor temp. = **145°F**
- Ambient air temperature **88°F**
- Air vent (duct) **49°F**
- Oil circulation, 3% by weight of refrigerant



What is standard J2210 versus J2788

What these standards are

- Two SAE standards for the air conditioning side of the automobile

What J2788 requires

- Recovery must reach 95 % in 30 minutes, without outside heat assistance
- No more than 0.5 oz of refrigerant left in the system



What does that mean on the technical side?

What it means for the machines

- They must be calibrated after a certain amount of use
- They must have a filter change system
- Purging must be automatic on this type of machine



Tooling, diagnosis, repair... A standard

What SAE requires of the shop

- Equipping itself with specific tools
- Following the recommendations for a compliant diagnosis and repair
- Respecting the safety measures

The standard concerned

- SAE J2845, for R1234yf and R744



The charging station

Mandatory standards

- SAE J-2843, or J-3030 for hybrid stations combining R134a and R1234yf
- The SAE number must be shown on the unit

What the station does

- Empties the system
- Recycles the refrigerant
- Charges the system



Features of the charging station under standards J-2843 and J-3030

Two versions

- Automatic
- Manual

What it performs

- Empties the system
- Recycles the recovered refrigerant
- Vacuum loss check, equivalent to a large leak test only
- Pressure check
- Charging



New feature

What the station does

- A pressure loss check of at least 5 minutes after pumping down
- The standard, however, treats it as a LARGE leak test only

What Québec requires on top of that

- The small leak check must be done by the technician
- With the known procedures, the nitrogen test for example
- Before performing a charge

⚠ WARNING The station only detects large leaks. In Québec, the small leak test remains the technician's responsibility, before any charging.



New feature

What the station does

- A negative pressure loss check
- Some stations show the vacuum on a scale in microns, which is far finer
- It can and must then perform a pressure test

Conversions to remember

- $30 \text{ inHg} = 762\,000 \text{ microns}$
- $1 \text{ inHg} = 25\,400 \text{ microns}$



What the pressure test consists of

The sequence

- After the vacuum decay test of at least 5 minutes, the unit moves on to the pressure test
- The station first charges the system to 15 % of its capacity
- The technician confirms the evaporator is leak free with an approved leak detector
- If it is leak free, the technician lets the station complete the remaining 85 %



Servicing the machine

Always keep a spare filter-drier on hand

The filter-drier

- It captures acid and particles, and removes water from the refrigerant
- Checking it reveals internal leaks, which happen regularly over time
- To be replaced after 68 kg (150 lb) of refrigerant filtered
- Unit warning at 45 kg (100 lb) of capacity used
- Unit lockout at 68 kg (150 lb): it stops working

The vacuum pump

- Change its oil after 10 hours of operation
- Beyond 10 hours without a change, the unit displays a message

Questions & Comments



Thank you all for taking part!

TRAINERS: Wilson Almeida
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